Statistical Analysis

Nigel Jhuen Rong Ng
efynn6@nottingham.edu.my
20516416

Kai Zheng Kan
szykk1@nottingham.edu.my
20613147

Dhia Aldeena Wibawa
efydw5@nottingham.edu.my
20497262

Jun Yang Chew
hfyjc21@nottingham.edu.my
20508940

Abstract

In this statistical analysis on the firm's transactional records on its beverages across stores in Selangor from 2023 to 2024, we seek to investigate the joint significance of historical depth of discount in daily sales as variables within a panel linear model. Our research aims to primarily investigate the possibility of a common factor within the panel linear model framework that serves as a driver to a spurious Granger causation of the time series of discount. As stores within the same geographic region are often subjected to common shocks, such as local demand conditions, synchronised promotional campaigns, or regional economic trends, it is reasonable that these phenomena may be encoded in the aforementioned common factor terms in the statistical models. We acknowledge the inflation in the dimension of the parameter space when estimating several seasonal dummies pertaining to seasonality and structural breaks of individual panels. We propose the usage of ensemble machine learning methods and a hybrid framework to capture these components in the data generating processes.

Keywords: Time Series, Microeconomics, Structural Breaks, Common Correlated Effects, Granger Causality, Ensemble Machine Learning

Contents

1 Introduction

Understanding the relationship between pricing strategies and revenue dynamics is a central concern in retail and consumer analytics. In particular, promotional activities such as discounts play a significant role influencing consumer purchasing behaviour and overall sales performance [1].

While discounts can increase sales in the short run, their broader impact remains uncertain. They may lead to inter-temporal substitution, where consumers shift purchases forward in time, potentially reducing future revenue through cannibalisation [2]. As a result, the net effect of discount strategies on aggregate revenue is not always positive and requires rigorous empirical analysis.

The effectiveness of these strategies often varies across distinct entities—such as stores or product categories—introducing unobserved heterogeneity that must be addressed within a panel data framework. Furthermore, retail datasets frequently exhibit cross-sectional dependence, as stores do not operate in isolation. Stores within the same geographical region are often exposed to common shocks, including local demand conditions, synchronised promotional campaigns, and regional economic trends, which can generate correlation across observational units [3]. In addition, spatial proximity may reinforce these correlations through regional spillovers and shared economic environments [4]. Consequently, revenue and discount patterns across stores are likely correlated via a latent common factor signal, necessitating a model that accounts for these interdependencies to avoid biased estimates.

To address these complexities, this work aims to estimate panel-wise time series regression models that address the possibly heterogenous dynamics of the revenue across each entity, whether store or drink, and also approximate a common correlated effects (CCE) model that purges latent common factors that may result in a spurious marginal correlation of the historical discounts, through common factors as confounding variables. We also wish to propose machine learning frameworks that capture individual responses to structural breaks and seasons, in which introducing individual specific dummy variables harm the estimation of the models themselves.

2 Methodology

2.1 Initial Observations in Data

Our dataset consists of tabulated daily records of sales of beverages belonging to various sub-categories, stores in various cities, and miscellaneous qualities such as the temperature of drink, type of bean/milk, and the venue of sale. Additionally, each sale record is tied to a discount included in the sale. These qualities of the data allow us to decide between aggregation strategies, to obtain panel datasets. For example, a store-wise panel total revenue dataset can be obtained by sum-aggregating revenue and discount and pivoting the original dataset by its store code, likewise for drink-wise panels via their product name.

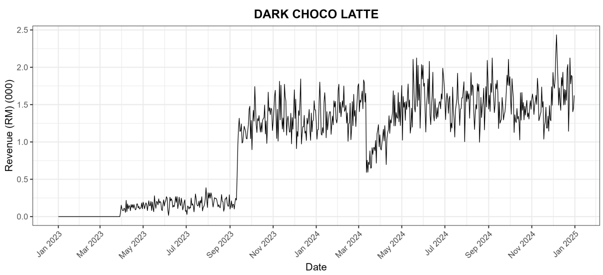
Hence, performing such data pre-processing steps to obtain the suitable format of panel data is necessary before fitting the data into time series models and statistical tests that pertain to the aforementioned research questions.

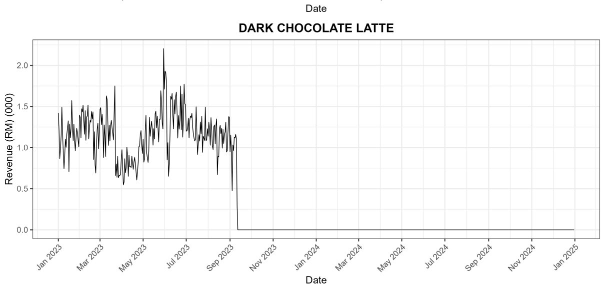
In the Exploratory Data Analysis (EDA) of our research, we come across several patterns in the dataset which may harm the final fit of the statistical model, and/or may be cleaned in such a way due to an underlying reason that we see fit. The below subsections are dedicated to explain these encounters. Note that for brevity, the store codes have been encoded as integers. The one-one map is included in the appendices.

In the following sections of the report, the dependent/target variable refers to the revenue generated per time period.

Dark Choco Latte → Dark Chocolate Latte

The following time series plots are of the drink-aggregated-revenue generated for the drinks “Dark Choco Latte” and “Dark Chocolate Latte”.





In our research, we identified this discrepancy within the dataset which, when directly fit into a dynamical model, will certainly cause a poor fit when attempting to fit to data with portions that have no observed dynamics. We tie this discrepancy to the similar names of both drinks; we assume in our research that, the drink was initially launched under the name “Dark Chocolate Latte”. After September 12, 2023, all stores have renamed the drink to “Dark Choco Latte”. Hence, the zero revenue generated by “Dark Chocolate Latte”. The above plots also give us an additional perspective that, there was a soft-launch period of the drink in which it is sold under the name “Dark Choco Latte”. However, for ease of modelling, we will rename “ Dark Choco Latte“ to “Dark Chocolate Latte” in the given dataset, effectively aggregating the total revenue generated by both drink names.

Unusual or unexplainable closing times

When inspecting the plots of store-aggregated revenue, we note the presence of time periods in which select few stores do not generate revenue at all. The fact that this pattern only arises in only a few stores make them unusual amongst other stores, but their cause still explainable. For example, closing weekends are the best guess as to why some stores do not generate any revenue at all during weekends.

Some stores, on the other hand, have longer runs of zero revenue generated. This may motivate us to include intercept shift dummies into statistical models we will fit, though, our deeper exploratory data analysis should justify why this should, or should not be done.

Negative selling-price records

Records in the dataset which were recorded to have a negative selling-price were dropped from the dataset.

2.2 Exploratory Data Analysis

Furthermore, our datasets aggregated into panels each consist of at least 20,000 instances of revenue, and several other non-engineered features, such as the City (store panel) / Product sub-category (drink panel), discount, and engineered features, such as the lagged revenue and discount, and rolling (window) mean statistics.

Structural Break Analysis

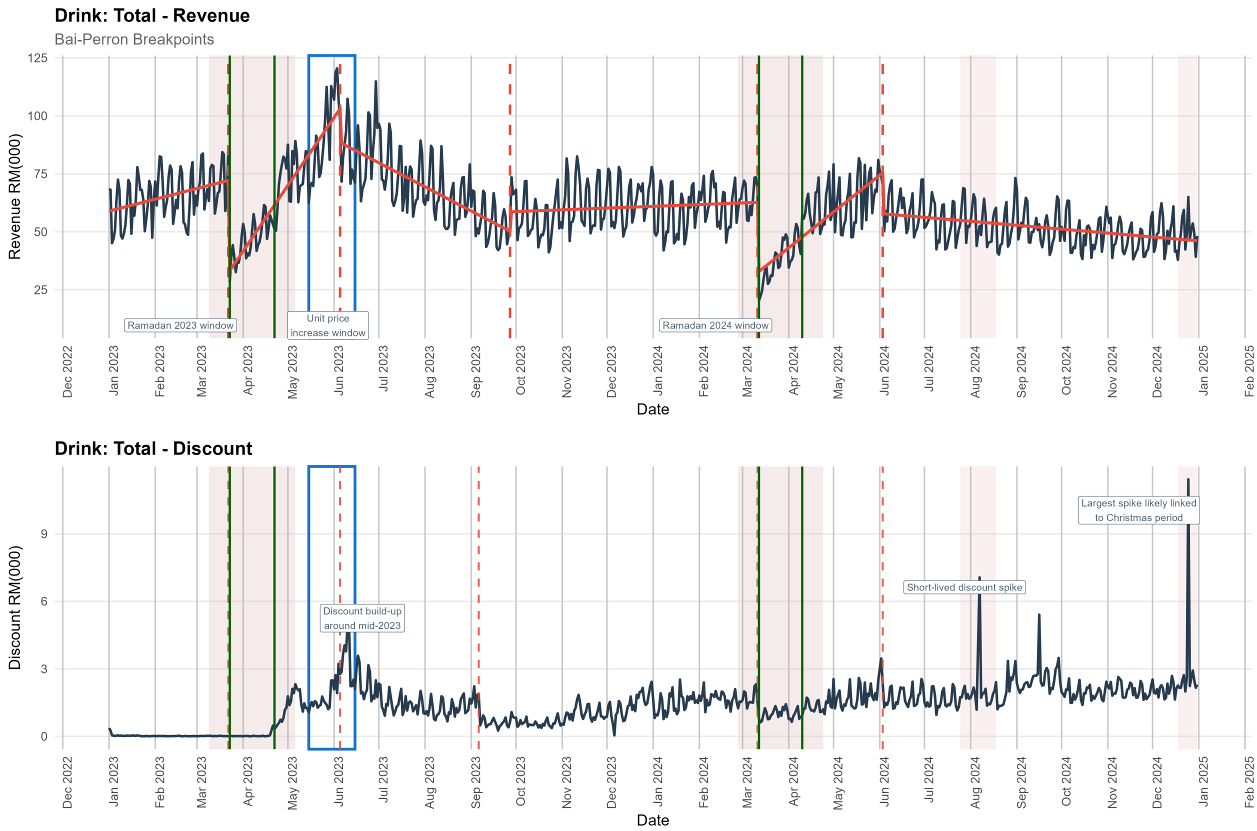


Figure 3: Time series plot of the total revenue across all panels with structural breaks and regimes indicated (top), accompanied with discount time series plot (bottom)

Table 1: Breakpoint dates

Breakpoint	Date
(1)	Mar. 22 '23
(2)	Jun. 5 '23
(3)	Sept. 27 '23
(4)	Mar. 3 '24
(5)	Jun. 3 '24

Figure 4 presents a time series plot of total revenue aggregated across all panels, together with Bai & Perron structural breakpoints [5]. The red dashed lines indicate structural breakpoints, separating the time series into distinct trend regimes indicated by the red lines. In the bottom plot, the indicated boxed region is a neighbourhood where the average selling prices of products faced an increase. The highlighted region in 7 August and 25 December 2024 correspond to spikes in the discount series. The latter date clearly corresponds to a promotional event on Christmas Eve 2024, while the former does not correspond to a specific public holiday, besides the beginning of August 2024.

An initial examination of the total series as well as individual store and drink panels reveals that revenue does not evolve smoothly over time but instead shifts between different regimes, indicating structural changes in the data generating process (DGP) of the revenue series.

Business Insights

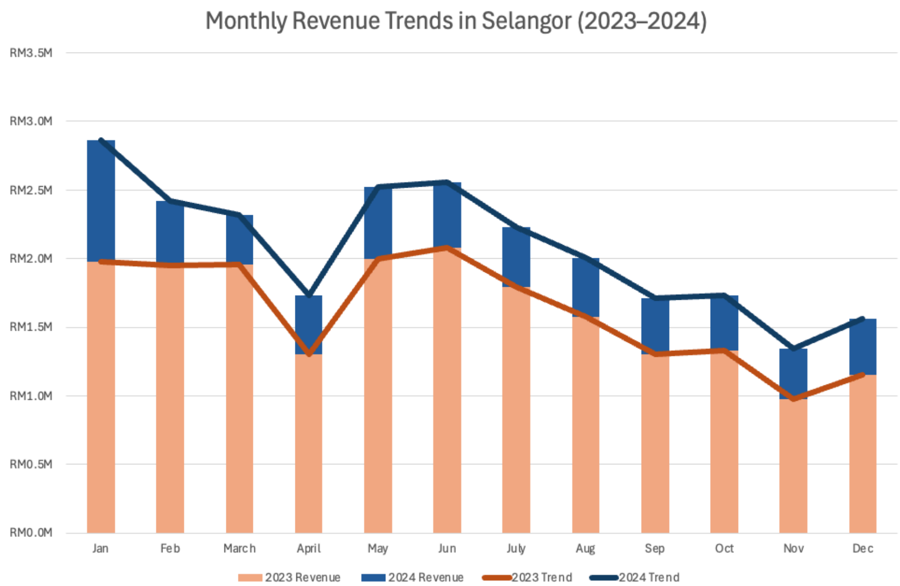


Figure 4: Stacked bar plot of '23 and '24 Revenue

In early 2023, revenue remains relatively stable and gradually increasing until the first structural break on 22 March 2023, after which a sharp decline is observed. This corresponds to the Ramadan period and reflects a negative intercept shift, where revenue abruptly transitions to a lower level. The timing suggests that the decline is linked to changes in consumer behaviour during the Ramadan period, such as reduced daytime consumption while fasting.

Following this decline, revenue recovers strongly to pre-festive levels at the end of Ramadan ‘23, and surpassing those levels in the weeks after. After this, total revenue levels reached a peak at June 2023, corresponding to the second structural break on 5 June 2023, which also marks the beginning of a declining regime.

The structural break on 27 September 2023 marks another transition to a slow growth regime. Unlike the downward shift due to Ramadan ‘23, this shift reflects a more gradual decline, suggesting this pattern is reflected in Figure 4 through multiple structural breakpoints, where revenue shifts step-by-step into lower levels, indicating a sustained weakening in revenue dynamics.

Similar regime changes are observed in 2024, with additional breaks identified around 3 March 2024 and 3 June 2024, indicating further adjustments in the underlying revenue level. Overall, these structural breaks highlight that the revenue process experiences multiple regime shifts across the two-year period. Figure 4 presents a stacked bar comparison of monthly revenue across 2023 and 2024, illustrating the relative contribution of each year within each month. It is evident that 2023 consistently contributes a larger share of total revenue than 2024 across all months, indicating that overall revenue levels in 2023 were substantially higher.

By comparing both line plots, the yearly business cycles of this firm observed on a monthly basis in 2023 and 2024 were quite comparable, with a structural dip in monthly revenue in April, then a rise to a local maximum in June, and finally a declining regime in the subsequent months, that is Q3-Q4 of the year, and a smaller dip in November. Therefore, the dynamics of the total revenue generated by the firm does not differ much between each year, apart from beginning at a lower level in the subsequent year due to the decline during the second half of the previous year.

To ensure comparability across transactions with varying quantities, the average selling prices per day was computed as the ratio of total selling price to quantity sold for each transaction. For brevity, this quantity is referred to as the unit price:

$$U_{i,t} = \text{Unit Price of Drink}_i \text{ at time } t = \frac{\sum_{j\in\mathcal{J}} \text{Price of } j\text{-th Drink}_i \text{ at time } t}{|\mathcal{J}_{i,t}|} \tag{1}$$

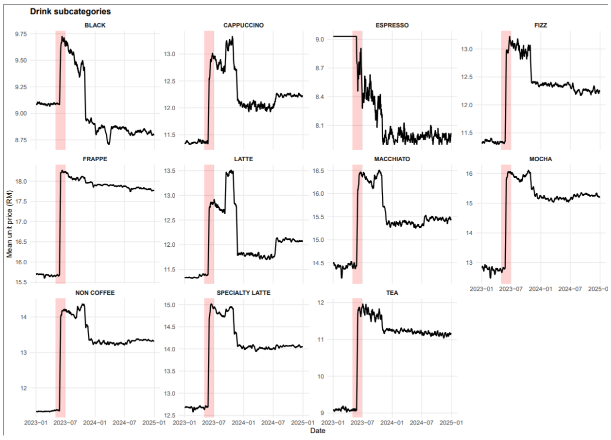


Figure 5: Individual plot of $U_{i,t}$

$\mathcal{J}_{i,t}$ is the index set of all instances of Drink$_i$ sold at time t. The same concept can be applied to some sub-category of drink, instead of computing the quantity at the drink level.

Inspection of the plots of average unit prices across individual beverage categories reveals a consistent and notable pattern. Across all sub-categories of products besides Espresso, which consist of only the Espresso drink itself, a sharp increase in unit prices is observed around late June 2023 — this period of time (May 1 - June 30) is highlighted in the plots in Figure 5.

This increase appears to be systematic across products suggesting a coordinated pricing adjustment rather than isolated prices changes. Following this adjustment, unit prices generally remain stable at a higher level throughout the remainder of 2023 and into 2024, with only minor fluctuations. There is no evidence of significant price reductions in 2024, indicating that the business maintained its revised pricing strategy. Thus, while it appears convenient to tie the structural break (2) in 2023 to the large positive shock in $U_{i,t}$, the same reason does not hold in 2024, when $U_{i,t}$ remains approximately level throughout 2024.

Research Motivation

Inspecting the total quantity of beverages sold by the firm informs us that the quantity sold follow similar dynamics. That is, the time series of quantity sold follow the same structural breaks and time regime shapes. In this work, we are interested to present the relevant statistical tests and models that incorporate the discount series as an exogenous variable in the models for revenue generated at the drink level, and also the store level.

By observing the bottom plot of Figure 3, it is apparent that, after Ramadan '23, the firm begins to incorporate a more aggressive promotional pricing strategy. This might entail a promotional event during this period, especially during June 2023, when the volume of discount distributed peaks around then. This might be naively related to the fact that $\uparrow$ Quantity of Drinks sold = $\uparrow$ Total depth of discounts tied to sales. However, this may not necessarily be the case, when customer demand for the firm's beverages is driven up without an exact promotional event in place. For example, marketing campaigns which may drive in potential customers. Therefore, in modelling the volume of revenue generated at a store/drink level, the discount series, in the respective cases, may not, or may be an explanatory variable in capturing the DGP of revenue at the store/drink level.

Moreover, it is interesting to note that although the product line of the firm experienced major shifts in the pricing in only 2023, the monthly dynamics of 2023 and 2024 still remained similar, with the 2024 series starting at a lower level being the only exception. As we have mentioned, if the unit price shift had a causal effect on the trend regime shift at the mid-point of the year 2023, the same causation cannot be said for the year 2024. By inspecting the individual plots of the revenue time series of each store and drink, similar structural breaks and shapes of regimes may also be found, albeit not consistent to the date. Therefore, we hypothesise that global shocks that drive the individual revenue at the drink and store level.

The application of Machine Learning (ML) models in this work tends to the inconsistency of structural breaks at the store and drink level. Although visually, individual stores/drinks exhibit fairly similar structural breaks, these breakpoints at the store/drink level are not precise and consistent to the breakpoints as listed in Table 1. This leaves the statistical modelling within the proposed framework in this research with two options, which are to either define a global set of structural breaks in which the level-specific model will incorporate as dummy variables, for all levels, or to define level-specific dummy variables to capture these structural breaks. In the former option, defining a global set of structural breaks enforces that the individual (level) dynamics structurally shift at the same time. In the latter option, defining individual dummy variables will inflate the dimension of the model space. Assuming N individuals and modelling the dynamics purely using b individual-specific structural breaks will yield $N \times b$ parameters alone.

Furthermore, conducting individual Bai–Perron breakpoint tests for each unit introduces additional statistical concerns. Breakpoint estimates are subject to sampling variability, and discrepancies across units may reflect estimation noise rather than true structural differences. This complicates inference and may lead to an over-fragmented representation of regime changes. Therefore, we rely on the ML methods to capture individual specific structural breaks and seasonality. By feature engineering lag terms into the input matrix of the ML model, we enable the model to learn temporal dependencies, including seasonality, without explicit encoding of the seasonal dummies into the input at all, provided that the lag terms capture one seasonal cycle.

Our research also examines the impact and complementarity of statistical frameworks to ML methods presented within this report. Despite advances in eXplainable Artificial Intelligence, ML methods struggle in reporting an interpretable framework as opposed to statistical time series models. ML methods lack the backbone of inferential tests to justify the specificity of the framework, and to validate underlying assumptions. However, if we define a framework to accept the statistical models presented in this report as a back-end to explain the dynamical component of our data, whilst leaving out the considerations of structural breaks and seasonality to a second ML stage, we gain the benefits of both the inferences of a statistical model, and the flexibility of ML methods in capturing the inconsistent structural breaks. The motivation behind this is that we wish to also examine the trade-off in complexity of the ML model in terms of architecture-specific hyper-parameters, such as the number of dense layers in a Deep Neural Network, number of trees in a Gradient Boosted Decision Tree, within a two-stage framework as proposed, over a full-stack ML model. Intuitively, the statistical model within a hybrid framework simplifies the data to be passed to the ML model, and hence the ML model can approximate the remaining component in the DGP well.

2.3 Econometrical Models

In our research, we employ variable coefficient panel linear models (VCM) of the form:

$$y_{i,t} = \alpha_i + \sum_{k=1}^{K} \gamma_i^{(k)} y_{i,t-k} + \sum_{q=0}^{Q} \beta_i^{(q)} x_{i,t-q} + \varepsilon_{i,t} \tag{2}$$

The notations included in this model and subsequent formulae are attached in Appendix A. Hence, within the context of this study, the model above approximates the DGP of the revenue of panel i, denoted by $y_{i,t}$, via K of its lags, Q lags of the depth of discount offered on the i-th panel, as well as its series added contemporaneously.

Prior lag selection

In the panel linear model stated in Equation 2, lag selection must be carefully made to ensure that the residuals produced final model fit does not have un-captured variances or correlation in the data. Initial observations in Auto-Correlation Function (ACF) plots of individual panel's revenue series indicate that the series itself is highly correlated with its lagged terms. We keep the selection of lags capped at $K, Q = 7$.

The Maddala-Wu test [6], [7] is a panel time series Fisher-type stationarity test that combines the individual one-sided Augmented Dickey-Fuller (ADF) tests each with the null $H_0 : \rho = 1$ on each panel, which indicates the presence of a unit root. That is, the p_i obtained in each ADF test is the p-value of the hypothesis test of $\rho = 1$ from Equation 3 on the i-th panel, in which $u_{i,t}$ is an ARMA(p, q) process, $p, q \geq 1$. That is, $\pi(B) u_{i,t} = v_{i,t}$, $v_{i,t} \sim \mathrm{WN}(0, \sigma_i^2)$, and $\pi(B) = 1 - \pi_1 B - \pi_2 B^2 + ... + \pi_{m-1} B^{m-1}$.

$$y_{i,t} = \rho y_{i,t-1} + \sum_{j=1}^{\infty} \pi_j \Delta y_{t-j} + v_{i,t} \tag{3}$$

$$y_{i,t} = \rho y_{i,t-1} + \sum_{j=1}^{m_i - 1} \pi_j \Delta y_{t-j} + v_{i,t}^{(m_i)} \tag{4}$$

$$P = -2 \sum_{i=1}^{N} \ln(p_i) \tag{5}$$

By the steps described in [8], Equation 3 is approximated by the regression in Equation 4, in which smallest m is selected such that autocorrelation is removed. It can also be selected by the Akaike Information Criterion (AIC) of regression. Hence, $p_i = P\left(t_i < t_{T_i - m; \alpha}\right)$, where $t_i = \dfrac{\hat{\rho}}{\mathrm{SE}(\hat{\rho})} \sim t_{T_i - m}$, in which $\mathrm{SE}(\hat{\rho})$ is calculated as in linear regression. Equation 5 gives the final p-value reported by the Maddala-Wu test, in which $P \sim \chi^2(2N)$.

We run the Maddala-Wu Fisher-type panel unit root test on the residuals of the models in the form in Equation 2, to verify that the individual panels are stationary. We use the Maddala-Wu test here as a diagnostic tool for model adequacy, helping us test for non-stationarity and also serial correlation. If the Maddala-Wu test fails to reject the null hypothesis that all panels' residuals are non-stationary, this calls for methods such as differencing the original series. Otherwise, rejection of the null calls for us to inspect the individual unit-root tests conducted under Maddala-Wu. An ideal result that indicates differencing is not required, as discernible from Equation 2, is that the residuals yielded by the said model are that individual panels' residuals are free from unit-roots while using $m = 0$. While smaller values of m suggest that only minimal correction for serial correlation is required in the unit-root test, the primary criterion for adequacy of the models in which the Maddala-Wu test reports is the stationarity of the residuals.

Verifying the heterogeneity of panels

Hsiao Test relies on a generalised F-test comparing the Residual Sum of Squares (RSS) of restricted against unrestricted models:

$$F = \frac{(RSS_{\text{Restricted}} - RSS_{\text{Unrestricted}}) / \text{ df1}}{RSS_{\text{Unrestricted}} / \text{ df2}} \tag{6}$$

,

$$RSS = \sum_{i=1}^{N} \sum_{t=1}^{T_i} \hat{u}_{it}^2 \tag{7}$$

Where df1 and df2 are the number of restrictions imposed under H_0 and in the unrestricted model respectively. For a model with N cross-sectional units, T_i time periods each panel, and K regressors, the RSS for each specification is computed as in Equation 7, where $\hat{u}_{it} = y_{it} - \hat{y}_{it}$ denotes the residual from the respective restricted or unrestricted model.

Table 2: Hsiao Test Hypotheses and Unbalanced Degrees of Freedom (N = Cross-sections, K = Explanatory variables, n = Total observations)

Hypothesis	**Null (H_0)**	**Restricted Model**	**Unrestricted Model**	**df1**	**df2**
H1	Overall Homogeneity (Common intercepts & slopes)	Pooled OLS (RSS_{Pooled})	Heterogeneous OLS (RSS_{Hetero})	$(N-1)(K+1)$	$n - N(K+1)$
H2	Homogeneous Slopes (Heterogeneous intercepts)	Fixed Effects (RSS_{FE})	Heterogeneous OLS (RSS_{Hetero})	$(N-1)K$	$n - N(K+1)$
H3	Homogeneous Intercepts & Slopes	Pooled OLS (RSS_{Pooled})	Fixed Effects (RSS_{FE})	$N-1$	$n-(N+K)$

Store Panel

The Store panel uses an expanded specification of $K = 15$ explanatory variables (1 contemporaneous discount + 7 discount lags + 7 revenue lags), with $N = 37$ cross-sections.

Table 3: Hsiao Test Results — Store Panel ($N = 37$, $K = 15$)

Hypothesis	**F-statistic**	**df1**	**df2**	**p-value**
H1	4.0188	576	26196	< 0.001
H2	3.9841	540	26196	< 0.001
H3	4.2811	36	26736	< 0.001

Drink Panel

The Drink panel evaluates $N = 37$ products with $K = 15$ explanatory variables (1 contemporaneous discount + 7 revenue lags + 7 discount lags).

Table 4: Hsiao Test Results — Drink Panel ($N = 37$, $K = 15$)

Hypothesis	F-statistic	df1	df2	p-value
H_1	2.2507	576	26196	< 0.001
H_2	1.7672	540	26196	< 0.001
H_3	9.3581	36	26736	< 0.001

All six hypotheses are rejected at the 0.1% significance level across both panels. The rejection of H1 confirms that a single pooled model is inappropriate for either dataset. More critically, the rejection of H2 establishes that slope coefficients differ across units — individual products respond differently to discounts, and consumer responsiveness to store-wide promotions varies significantly by geography. The rejection of H3 further confirms that baseline revenue levels differ meaningfully across units.

The `plm` package in R natively computes degrees of freedom assuming a balanced grid ($N \times T$), which does not strictly hold for unbalanced panels. Since the inclusion of 7 lags truncates the effective time index from 731 to 724 days, the software arrives at a theoretical $n = 37 \times 724 = 26,788$ for both panels. For the Store panel, the true observation count is $n = 22,829$, reflecting the unbalanced nature of store-level trading records. For the Drink panel, the true and theoretical counts are approximately aligned given the more complete product-level coverage.

Despite this computational discrepancy, the core inference remains valid and asymptotically robust. The foundation of the F-statistic rests on the residual sums of squares, which the software computes accurately from the true unbalanced dataset via $\text{RSS} = \sum_{i=1}^{N} \sum_{t=1}^{T_i} \hat{u}_{it}^2$. The degrees of freedom serve merely as scaling constants. While substituting the theoretical df2 for the true df2 slightly distorts the denominator's finite-sample scaling, standard large-sample econometric theory demonstrates that this distortion is mathematically inconsequential.

$$F = \frac{\chi_J^2 / J}{\chi_{n-k}^2 / (n-k)} \tag{8}$$

We may leverage lemmas and theorems in econometrics as detailed in Appendix C to formally prove the asymptotic behaviour of the F-statistics in this test. By definition, an F-distributed random variable is the ratio of two independent Chi-square variables, each scaled by their respective degrees of freedom given in Equation 8. Multiplying both sides of Equation 8 by the number of restrictions (J) gives the random variable $J \times F$, in which by Corollary 5.0.3.1, we have that $J \times F \xrightarrow{d} \chi_J^2$

Therefore, once the denominator degrees of freedom exceed approximately 1,000, this asymptotic convergence is effectively complete, and the F-distribution stabilises entirely into a scaled Chi-square distribution. Because both the true df2 (22,829) and the theoretical df2 (26,788) are effectively infinite in this context, their respective denominators have both mathematically converged to 1. Consequently, they share identical critical thresholds on the asymptotic curve. As all reported F-statistics far exceed these thresholds at the 0.1% significance level, the degree-of-freedom discrepancy has no bearing on any of the conclusions drawn.

Collectively, these findings necessitate a fully heterogeneous estimator that permits both slopes and intercepts to vary freely across individual units.

Random Effects vs Fixed Effects

A panel linear model with individual fixed effects absorbs unit-specific intercepts α_i that are constant over time but vary across units. The random-effects model imposes the stricter assumption that these intercepts are uncorrelated with the explanatory variables. That is, $\mathbb{E}(\alpha_i \mid \boldsymbol{X}_i) = 0$. Treating α_i as random under these conditions would conflate the unit-specific heterogeneity with the slope estimates, inducing omitted-variable bias.

To test this formally, we employ Mundlak's [9] auxiliary regression. Rather than testing directly whether α_i from a homogeneous panel linear model is correlated with $\boldsymbol{X}_i$, Mundlak's formulation approximates $\mathbb{E}(\alpha_i \mid \boldsymbol{X}_i)$ as a linear function of the within-unit covariate means:

$$\alpha_i = \overline{\boldsymbol{x}}_i^\top \boldsymbol{a} + w_i, \quad w_i \sim \mathcal{N}(0, \sigma_w^2) \tag{9}$$

If the slope vector $\boldsymbol{a}$ is jointly zero, the group means carry no additional explanatory power and the random-effects specification is valid. A rejection of $\boldsymbol{a} = \boldsymbol{0}$ implies that the unobserved heterogeneity is systematically related to the regressors, rendering random effects inconsistent. The Mundlak regression yields a standard F-statistic that is straightforward to interpret and does not require estimating a variance-covariance matrix, unlike the Hausman test, so the former is preferred to evaluate the consistency of random vs fixed effects.

Applying this auxiliary regression to both the store and drink panels, we obtain $p < 0.05$ in the F -test for $\boldsymbol{a} = \boldsymbol{0}$ in both cases. The null is rejected, confirming that the unit-specific intercepts are correlated with the regressors. Economically, this implies that the fixed effects α_i absorb structural differences across units. Omitting these from the model would bias the slope estimates. We therefore adopt fixed effects throughout.

It should be noted that this conclusion operates within a homogeneous slope framework, as both the Mundlak test and the Hausman test assume that $\boldsymbol{\beta}_i = \boldsymbol{\beta}$ for all i. The Hsiao test, discussed in the preceding subsection, already established that this assumption is violated in our panels: slopes are heterogeneous across units. The fixed-effects decision nonetheless remains relevant for the intercept specification. More importantly, even after conditioning on unit fixed effects, stores and drinks within the same retail chain are exposed to common shocks — chain-wide promotional campaigns, supply disruptions, seasonal demand shifts — that induce cross-sectional dependence in the residuals. Standard fixed-effects estimators do not account for this dependence and will yield inconsistent estimates when unobserved common factors are present. This motivates the adoption of time series regression models that account for these common factors.

Granger Causality

Granger Causality answers the joint influence of the time series of each panel's historical volume of discount distributed to the revenue generated at time t within a panel linear model framework as in Equation 2. Our work also seeks to investigate whether this Granger causality is driven by individual panel dynamics, or by intrinsic common factors whose dynamics drive the Granger causality. We assess the contribution of the discount series under a framework modified from the form given in Equation 2, to Equation 10, by letting $Q = K$, without considering the contemporaneous regressor x. This form of panel linear model to assess Homogeneous Granger Non-Causality was proposed by [10]. When running the statistical tests, as well as fitting models, it is imperative to drop some panels that have tiny sample sizes — if not, some panels will render us unable to asymptotic results of statistical tests, nor have a high power of statistical tests themselves. Fortunately, this issue only arises in store-aggregated panels, with stores $29, 31, 38$.

We can write Equation 10 in vector form by defining the matrices $\boldsymbol{X}_i = \begin{bmatrix} \boldsymbol{x}_i^{(1)} & \boldsymbol{x}_i^{(2)} & \cdots & \boldsymbol{x}_i^{(K)} \end{bmatrix}$, $\boldsymbol{Y}_i = \begin{bmatrix} \boldsymbol{y}_i^{(1)} & \boldsymbol{y}_i^{(2)} & \cdots & \boldsymbol{y}_i^{(K)} \end{bmatrix}$, and finally $\boldsymbol{Z}_i = [\mathbf{1} \quad \boldsymbol{X}_i \quad \boldsymbol{Y}_i]$. Then, the Wald test statistic for individual Granger causality test is formulated in Equation 13. The test statistic W^{HNC} is of the Homogeneous Non-Causality (HNC) test of the joint contribution of the regressors x to y. Thus, the null hypothesis is that $\beta_i^{(k)} = 0, \quad \forall k \in \{1, 2, \cdots, K\}$, and the alternative is that $\beta_i^{(k)} \neq 0$ for some lag order k, over all panels.

$$\begin{aligned} y_{i,t} = \alpha_i + \sum_{k=1}^{K} \gamma_i^{(k)} y_{i,t-k} \\ + \sum_{k=1}^{K} \beta_i^{(k)} x_{i,t-k} + \varepsilon_{i,t} \end{aligned} \tag{10}$$

$$\boldsymbol{\theta}_i := \begin{bmatrix} \alpha_i & \gamma_i^\top & \beta_i^\top \end{bmatrix}^\top \tag{11}$$

$$\boldsymbol{y}_i = \boldsymbol{Z}_i \boldsymbol{\theta}_i + \boldsymbol{\varepsilon}_i \tag{12}$$

A1: For each i , $\boldsymbol{\varepsilon}_i$ are all $\text{WN}(0, \sigma_{\varepsilon,i}^2 I)$
A2: Residuals across panels are independent. $\mathbb{E}(\varepsilon_{i,t}\varepsilon_{j,s}) = 0$.

$$\begin{aligned} W_i = \frac{\tilde{\boldsymbol{\varepsilon}}_i^\top \Phi_i \tilde{\boldsymbol{\varepsilon}}_i}{(T_i - 2K - 1)^{-1} \tilde{\boldsymbol{\varepsilon}}_i^\top M_i \tilde{\boldsymbol{\varepsilon}}_i}, \\ \forall i = 1, \cdots, N \text{ panels, where} \\ \tilde{\varepsilon}_i = \frac{\varepsilon_i}{\sigma_{\varepsilon,i}} \end{aligned} \tag{13}$$

$$\overline{W}^{\text{HNC}} = \frac{1}{N} \sum_{i=1}^{N} W_i \tag{14}$$

$$\widehat{Z_N} = \left(\frac{2K}{N}\right)^{-\frac{1}{2}} \left(\widehat{\overline{W}}_N^{\text{HNC}} - K\right) \xrightarrow{d} \mathcal{N}(0,1) \tag{15}$$

$\boldsymbol{\Phi}_i$ and $\boldsymbol{M}_i$ are positive semi-definite (PSD) projection matrices.[1] Therefore, given the core assumptions on ε_i of HNC which were verified by inspecting diagnostic Q-Q plots of residuals from each panel, the estimated residuals of the model is given by $\hat{\boldsymbol{\varepsilon}}_i$. While the zero-mean assumption of **A1** was satisfied, a limitation is that some serial correlation may still be present for smaller lag orders and the estimates for W_i may be slightly optimistic. By **A1** and Slutsky's Theorem:

$$\widehat{W}_i = \frac{\tilde{\varepsilon}_i^\top \boldsymbol{\Phi}_i \tilde{\varepsilon}_i}{(T_i - 2K - 1)^{-1} \tilde{\varepsilon}_i^\top M_i \tilde{\varepsilon}_i} \xrightarrow{p} \tilde{\varepsilon}_i^\top \boldsymbol{\Phi}_i \tilde{\varepsilon}_i \tag{16}$$

in which the numerator is a sample estimate from a $\chi^2(K)$ distribution, i.e., $\mathbb{E}(W_i) = K$ and $\mathbb{E}(W_i^2) = 2K + K^2$. $\overline{W}_i$ is the average statistic, with (a) $\mathbb{E}\left(\overline{W}_i\right) = K$ and (b) $\mathbb{V}ar\left(\overline{W}_i\right) = 2K/N$, in which **Assumption A2 holds.** Where this assumption is violated, the asymptotic variance (b) is underestimated and lacks an additive covariance term. The estimated Z-statistic for HNC is hence given by Equation 15.

We use this asymptotic result with large N, under the assumption that individual T_i are sufficiently large. Specifically, $T_i \gg 5 + 2K$ in this work. [10] supports this, whose authors mention that, although individual $W_i \sim \chi^2_{T_i - 2K - 1}$ are not identically distributed for varying T_i, the moments do converge to those of W_i. These details are included in the appendices.

[1] symmetric and idempotent

Thus, we run the HNC test under the specifications above, for both store panels and drink panels. The obtain Wald test statistics for several lag orders L of up to 7 have been tabulated. For both the store and drink panel model, the data serves as significant evidence in favour of the alternative that $\beta_i^{(k)} \neq 0$ for some $k \in \{1, 2, ..., K\}$, $\forall L \in \{1, ..., 7\}$ at an $\alpha = 0.001$ significance level.

The computed pairwise correlation between the panels at the lag orders $K \in \{1, ..., 7\}$, as tabulated in Table 5 and Table 6, were large and we suspect that these coefficients indicate a non-negligible effect on the computed test statistics.

The presence of cross-sectional dependence (CSD) as indicated by the fair correlation coefficient average between panels, indicates that the results of HNC are not directly indicative of the Granger-causality of discount to revenue at the panel level. This is because, $\sigma^2_{\varepsilon,i}$ in the first definition of W_i in Equation 13 is approximated via $\sigma^2_{\varepsilon,i} \approx (T_i - 2K - 1)^{-1} \hat{\boldsymbol{\varepsilon}}_i^\top \hat{\boldsymbol{\varepsilon}}_i$, when in reality $\sigma^2_{\varepsilon,i} = (T_i - 2K - 1)^{-1} \hat{\boldsymbol{\varepsilon}}_i^\top \hat{\boldsymbol{\varepsilon}}_i + 2 \sum_{i<j} \mathbb{C}ov\left(\hat{\boldsymbol{\varepsilon}}_i, \hat{\boldsymbol{\varepsilon}}_j\right)$. And as the measures of CSD, ρ, as in the previous paragraph, is large, so is $\mathbb{C}ov\left(\hat{\boldsymbol{\varepsilon}}_i, \hat{\boldsymbol{\varepsilon}}_j\right)$, leading to an under-approximation of $\sigma_{\varepsilon,i}$. Thus, W_i is inflated, and p-values are deceivingly significant.

$$\boldsymbol{y}_i = \boldsymbol{Z}_i \boldsymbol{\theta}_i + \boldsymbol{\Gamma}_i^\top \boldsymbol{f}_t + \boldsymbol{u}_i \tag{17}$$

$$\tilde{\boldsymbol{y}}_i = \tilde{\boldsymbol{Z}}_i \boldsymbol{\theta}_i + \tilde{\boldsymbol{u}}_i \tag{18}$$

Hence, in our research, we believe that there exists some driving force common throughout all panels which simultaneously gives rise to CSD and produces optimistic HNC test results. This is described as an unobserved common factor $\boldsymbol{f}_t$ in the residuals by the authors of [11]. In vector form, the model in Equation 10 in vector form is given by Equation 17, where $\boldsymbol{u}_i$ is not necessarily a zero mean white noise — idiosyncratic — due to CSD. We define the projection matrices $\overline{\boldsymbol{Z}}_i := \begin{bmatrix} \boldsymbol{1} & \overline{\boldsymbol{X}} & \overline{\boldsymbol{Y}} \end{bmatrix}$, $\boldsymbol{M}_{\overline{\boldsymbol{Z}}_i} := \boldsymbol{I}_{T_i} - \overline{\boldsymbol{Z}}_i \left(\overline{\boldsymbol{Z}}_i^\top \overline{\boldsymbol{Z}}_i\right)^{-1} \overline{\boldsymbol{Z}}_i^\top$, then apply the projection $\boldsymbol{M}_{\overline{\boldsymbol{Z}}_i}$ on both sides of Equation 17 to obtain Equation 18, which is the Common Correlated Effects Mean Group Estimator model.

To control for this effect, we apply a projected variant of the variable coefficient model as in Equation 2, in which we define a suitable projection matrix to approximately zero out this common effect within the dependent signal $y_{i,t}$, and regressors $\boldsymbol{x}_i$. Consequently, running the HNC test on the projected variables will give us a clearer idea on the granger-causal relationship between the overall panels' volume of discount distributed to the revenue generated.

$$\hat{\beta}_i = \left(\boldsymbol{Z}_i^T \boldsymbol{M}_{\overline{Z}_i} \boldsymbol{Z}_i\right)^{-1} \boldsymbol{Z}_i^T \boldsymbol{M}_{\overline{Z}_i} \boldsymbol{y}_i \tag{19}$$

$$\hat{\beta}_{\text{MG}} = \frac{1}{N} \sum_{i=1}^{N} \hat{\beta}_i \tag{20}$$

In this projection, we account for the contemporaneous effect of the discount series, $x_{i,t}$, and lag terms of $y_{i,t}$ and $x_{i,t}$ up to $K = Q = 7$ lag orders. After we obtain the projected data, we will run the HNC tests under the variable coefficient framework as in Equation 10, whereby using $\tilde{\boldsymbol{y}}_i$ and $\tilde{\boldsymbol{x}}_i$. The coefficients of the model stated in Equation 18, $\boldsymbol{\beta}_i$, is estimated by Equation 19, and then the Mean Group (MG) coefficient estimator by Equation 20.

Table 5: Cross-Sectional Dependence Test — $\tilde{Z}$ statistic p-values and average correlation coefficients by lag order (Store)

	$\varepsilon_i = \Gamma_{i,t}^\top f_t + u_i$			$u_i \sim \mathrm{WN}(0, \sigma_i^2)$		
Lag Order	$\tilde{Z}$ (p-value)	$\overline{\rho}$	$\lvert\overline{\rho}\rvert$	$\tilde{Z}$ (p-value)	$\overline{\rho}$	$\lvert\overline{\rho}\rvert$
1	30.0490 (2.25×10^{-198})	0.2237	0.4515	6.2427 (4.30×10^{-10})	−0.0244	0.1459
2	32.7476 (3.29×10^{-235})	0.2283	0.4321	6.6179 (3.64×10^{-11})	−0.0232	0.1464
3	24.1174 (1.64×10^{-128})	0.2277	0.4314	4.1469 (3.37×10^{-5})	−0.0228	0.1456
4	19.2950 (5.92×10^{-83})	0.2295	0.4297	3.8049 (1.42×10^{-4})	−0.0225	0.1460
5	14.1732 (1.34×10^{-45})	0.2175	0.4266	4.3444 (1.40×10^{-5})	−0.0217	0.1453
6	7.5297 (5.09×10^{-14})	0.2012	0.3694	4.1736 (3.00×10^{-5})	−0.0227	0.1447
7	5.4253 (5.79×10^{-8})	0.1956	0.3297	4.0344 (5.47×10^{-5})	−0.0232	0.1330

Table 6: Cross-Sectional Dependence Test — $\tilde{Z}$ statistic p-values and average correlation coefficients by lag order (Drink)

	$\varepsilon_i = \Gamma_{i,t}^\top f_t + u_i$			$u_i \sim \mathrm{WN}(0, \sigma_i^2)$		
Lag Order	$\tilde{Z}$ (p-value)	$\overline{\rho}$	$\lvert\overline{\rho}\rvert$	$\tilde{Z}$ (p-value)	$\overline{\rho}$	$\lvert\overline{\rho}\rvert$
1	12.6441 (1.21×10^{-36})	0.3778	0.3795	5.2639 (1.41×10^{-7})	0.0289	0.0886
2	12.8107 (1.43×10^{-37})	0.3839	0.3857	3.1620 (1.57×10^{-3})	0.0320	0.0875
3	10.1566 (3.10×10^{-24})	0.3832	0.3851	2.9483 (3.20×10^{-3})	0.0329	0.0860
4	8.1314 (4.25×10^{-16})	0.3837	0.3857	2.4170 (1.56×10^{-2})	0.0333	0.0850
5	6.4203 (1.36×10^{-10})	0.3744	0.3763	2.3660 (1.80×10^{-2})	0.0335	0.0842
6	5.9506 (2.67×10^{-9})	0.3442	0.3449	2.8707 (4.10×10^{-3})	0.0340	0.0832
7	4.9171 (8.78×10^{-7})	0.3300	0.3305	2.6493 (8.07×10^{-3})	0.0344	0.0824

The p-values for HNC under all lag orders have notably decreased several orders of 10 after accounting for the common factor. Moreover, the cross-sectional correlation coefficients have decreased to an amount which we will deem fairly negligible in this research. A limitation of running these HNC tests is that the causality tests ran before and after accounting for common factors is that the contemporaneous effect of the exogenous series, $\boldsymbol{x_t}$, are not at all accounted for, and only indirectly accounted for within the projection in the latter case. Moreover, individual Wald statistics which help estimate $\tilde{Z}$ may be optimistic as a result of serial correlation in within-panel residuals. However, we can still conclude that, common factors had heavily inflated the individual test statistics to yield optimistic Granger-causality results.

This is because, the HNC tests are causality tests to investigate solely the influence of historical (lagged) exogenous signals towards the current revenue. Therefore, these tests do not necessarily serve as tell-tale signs of the joint significance of the presented historical signals under the framework Equation 2, where the index $q = 0$ is included. The valuable interpretations of these HNC tests, however, will be discussed in subsequent sections.

2.4 Ensemble Machine Learning

LightGBM, developed by Microsoft, has been detailed in [12] as an architecture of a gradient-boosted decision tree (GBDT) that introduces novel *Gradient-based One-Side Sampling* (GOSS), and *Exclusive Feature Bundling* (EFB) algorithms, with which algorithmically selects samples from the dataset based on gradient magnitudes, and dimensionality reduction by feature bundling, to tackle historical issues in efficiency and scalability of such machine learning (ML) models.

This is in line with the properties of our dataset on which we perform time series analysis, as well as our research questions — lag features and rolling mean statistics engrained in the input matrix of *LightGBM* are essential to model temporal dependencies, thus calling for the feature engineering on the panel dataset. With each feature engineered, the dimensionality of the feature space increments, and becomes problematic especially for smaller sized panels (60 samples) when its feature space becomes significantly more sparse; curse of dimensionality.

Furthermore, this work is motivated to expose the reduction, or not, in complexity of ML models under the assumption of a statistical back-end, in which the dataset $\mathcal{D}$ fed into *LightGBM* are the residuals of a statistical model, whose frameworks discussed in the previous section.

Smart sample (GOSS) and feature selection (EFB)

The primary benefit of leveraging the algorithms within the *LightGBM* framework, namely *GOSS* and *EFB*, over traditional Gradient Boosting Decision Tree (GBDT) methods is the reduction in computational complexity. Past GBDT methods rely on scanning the entire set of training data instances to derive a subset of split points, and then, estimating the information gain of all split points in said subset.

GOSS samples $100a\%$ proportion of the training instances with the top largest negative gradients, of which belong in the set A, then, randomly samples the remaining $|A^c| \times 100b\%$ proportion of the training instances in each training run. The remaining latter samples are rescaled by a factor $\frac{1-a}{b}$ in estimating the variance, so as to pad the sum of gradients to compute the mean of gradients in the left and right child notes, respectively.

By observing the significance of lag terms as far as 7 for each of the discount and revenue series within a panel linear model setting, we hypothesise that some features, even within a *LightGBM* framework, are insignificant during inference. *LightGBM*, specifically its *EFB* algorithm, enables us to test this hypothesis, in which exclusive features are bundled into a single feature, namely, an exclusive feature bundle. This achieves (a) sped up training process of a GBDT architecture, and (b) dimensionality reduction without sacrificing explanatory power.

Loss Function

Past research that have utilised *LightGBM* in forecasting applications [13] detail their choices of loss function by the nature of their respective datasets. They used the Pinball Loss function in the training step of their *LightGBM* models, which has a reduced sensitivity to outliers, unlike loss functions like Root Mean Squared Error (MSE), in which the outliers carry a heavier weight in computing the loss function. This is because, MSE grows quadratically w.r.t the error $y_i - \hat{y}_i$. In our hybrid modelling approach, since we wish for *LightGBM* to specifically capture panel specific structural breaks, whose large intercept shifts may be considered outliers, we may utilise the mentioned properties of RMSE loss as stated in Equation 21. Findings from our EDA allow us to make a more informed decision on the loss functions on which *LightGBM* is trained.

Results from past literature have made the LightGBM model particularly attractive in the line of time series analysis. For example, [13] highlighted key contributions in completing a comparative analysis of several established ML and econometrical baseline models. They concluded that *LightGBM* and its various configurations were highly effective in forecasting methods compared to econometrical applications. In our study, we would also like to present a feature contribution analysis on our dataset to highlight the significance of each engineered feature in the input matrix of *LightGBM*, as well as the panel indicators during key structural break dates in revenue. Hence, the research presented is able to demonstrate whether *LightGBM* trained and fine-tuned on the entire input matrix benefits significantly from such indicator variables, as a way to acknowledge the panel nature of the dataset.

$$\mathcal{L}(\boldsymbol{y}, \hat{\boldsymbol{y}}) = \sqrt{\frac{1}{\sum_i T_i} \sum_{i=1}^{N} \sum_{t=1}^{T_i} \left(y_{i,t} - \hat{y}_{i,t}\right)^2} \tag{21}$$

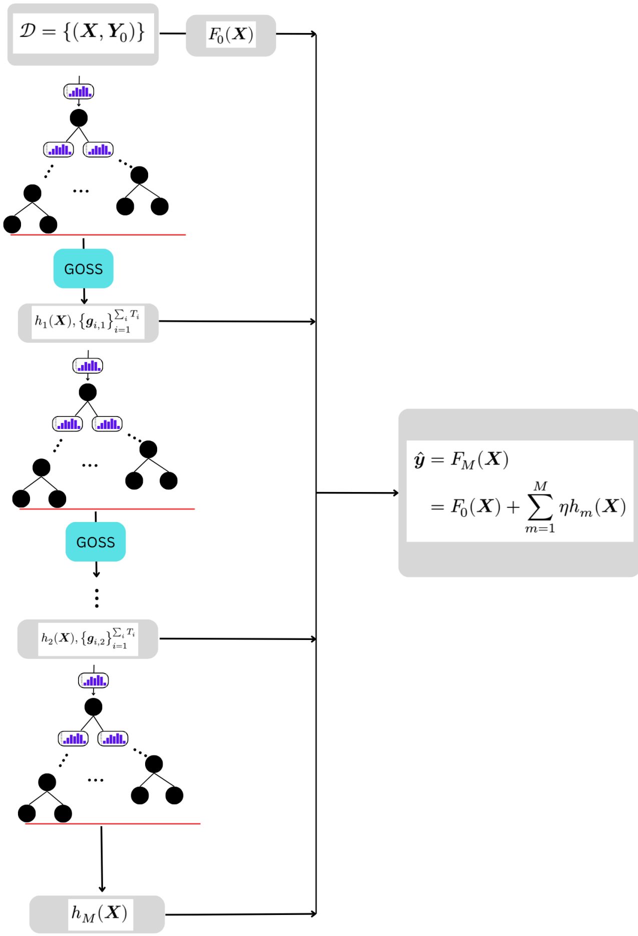


Figure 9: *LightGBM* architecture.

Variance Gain estimation, Histogram Binning, and node split method

Suppose *LightGBM* is trained on a dataset $\mathcal{D} = \{(\boldsymbol{X}, \boldsymbol{Y}_0)\}$, where $\boldsymbol{X}$ here are the features in the input matrix and $\boldsymbol{Y}_0$ are the continuous labels. Let $\boldsymbol{X} \in \mathcal{X}^p$, where $\mathcal{X}^p$ is a p-dimensional feature space. Hence, one singular instance $\boldsymbol{X}$ of all p-features can be represented by a p-dimensional vector.

These features however, are not directly fed into a single node for variance gain estimation — *EFB* is applied for an edge in computational efficiency in this GBDT method as a dimensionality reduction method to map $\boldsymbol{X}$ to a $\mathcal{X}^d$ feature space, where $d < p$.

$$\hat{V}_{j\,|\,O}(s) = \frac{1}{\sum_i T_i} \left(\frac{\left(\sum_{x_{ij} \le s} g(x_{ij})\right)^2}{n^j_{l|O}(s)} + \frac{\left(\sum_{x_{ij} > s} g(x_{ij})\right)^2}{n^j_{r|O}(s)} \right) \tag{22}$$

$$\hat{V}_j(s) = \frac{1}{\sum_i T_i}\left(\frac{\left(\sum_{x_{ij}\in A_l} g(x_{ij}) + \frac{1-a}{b}\sum_{x_{ij}\in B_l} g(x_{ij})\right)^2}{n_l^j(s)} + \frac{\left(\sum_{x_{ij}\in A_r} g(x_{ij}) + \frac{1-a}{b}\sum_{x_{ij}\in B_r} g(x_{ij})\right)^2}{n_r^j(s)}\right) \tag{23}$$

$A_l, B_l, A_r, B_r \subseteq O$. $A_l \coloneqq \{x \in A : x \leq s\}, A_r \coloneqq \{x \in A : x > s\}$, similarly for B_l, B_r. s is the split-point that is representative of one histogram bin of the binned set of all possible split points.

It is at variance gain estimation where the histogram-based algorithm trims the computational complexity of solving the optimal split. Whereby, consider the optimisation problem:

$$s^* = \arg\max_{(s,j)} V_j(s) = \arg\max_{j\in I(\mathcal{X}^d)} \left\{\max_{s\in H(S)} V_{j|O}(s)\right\}_{j\in I(\mathcal{X}^d)} \tag{24}$$

in which $V_j(d)$ from Equation 22 is estimated by Equation 23. The tuple (s, j) is lexicographically ordered[2] GBDT algorithms typically exhaust S instead of $H(S)$, where $H(S)$ is the histogram-binned possible split point set.

Regression with LightGBM

With the objectives of the algorithms that construct *LightGBM* detailed, the steps for *LightGBM* to perform continuous regression becomes more apparent. The process is as detailed in Figure 9.

1. *LightGBM* assumes the initial model $F_0(\boldsymbol{X}) = \overline{\boldsymbol{Y}_0}$. Iterative boosting begins in the subsequent steps.
2. h_1 is fit on the residual component, i.e, $\boldsymbol{Y}_0 - F_0(\boldsymbol{X})$ — optimal split points are computed in each child node until termination producing leaf nodes. Termination may occur due to the maximum depth reached and/or child nodes formed. GOSS samples are candidates
3. h_1 does not produce a perfect fit itself. Thus, regression component itself yields some residuals (negative gradients), namely, $\boldsymbol{g}_1$
4. Negative gradients trickle down to h_2
5. Steps $2 \rightarrow 4$ repeated, ending with the update step $F_m(\boldsymbol{X}) = F_{m-1}(\boldsymbol{X}) + \eta \cdot h_m(\boldsymbol{X})$ at each loop, creating a cascading till M boosting trees are created.

Hyper-parameter Bayesian Optimisation (Optuna)

In our research, we utilise an iterative Bayesian Optimisation (BO) framework to search the model space constructed by a high-dimensional (>2) grid of hyper-parameters to compute the most effective configuration of hyper-parameters, as well as optimal lag choice of the time series fed into *LightGBM*, namely, *Optuna*. Thus, in this work, *Optuna* effectively fine-tunes *LightGBM* according to the processed dataset, i.e., store-aggregated and drink-aggregated panels.

[2] In which s^* is computed for individual j, then maximised across all j.

Table 7: LightGBM model hyper-parameters used in this study.

Parameter	Search Grid	Description
`feature_fraction`	$[0.5, 1]$	Fraction of features used per tree
`max_depth`	[4, 5, ..., 16]	Maximum tree depth
`num_leaves`	$[\min(16, 2^{\texttt{max\_depth}-1}), 2^{\texttt{max\_depth}-1}]$	Maximum number of leaves per tree
`min_child_samples`	[10,20,...,50]	Minimum samples required in a leaf
`num_tree`	[log[50,...,250]]	Number of boosting iterations
`lambda_l2`	$[\log[10^{-8}, ..., 10]]$	l_2 regularisation term

Tree-structured Parzen Estimator (TPE) sampler was used within this framework to minimise the RMSE, in each trial, by selecting from a set of candidates in each trial by a probability of improvement criterion given by Equation 25. y^* is selected based on a splitting algorithm, where y^γ, the point which splits the top and bottom quantiles, will represent y^*. A quantity isomorphic to Equation 25 is $r(\boldsymbol{x} \mid \mathcal{C})$.

$$\mathbb{P}(y \leq y^* \mid \boldsymbol{x}, \mathcal{C}) = \int_{-\infty}^{y^*} p(y \mid \boldsymbol{x}, \mathcal{C}) dy \quad (25)$$

$$r(\boldsymbol{x} \mid \mathcal{C}) := \frac{p(\boldsymbol{x} \mid \mathcal{C}^{(l)})}{p(\boldsymbol{x} \mid \mathcal{C}^{(u)})} \overset{\text{rank}}{\simeq} \mathbb{P}(y \leq y^\gamma \mid \boldsymbol{x}, \mathcal{C})$$

$\boldsymbol{x}$ here is a candidate vector of continuous hyper-parameters. $p(y \mid \boldsymbol{x}, \mathcal{C})$ is estimated by first estimating $p\big(y \mid \boldsymbol{x}, \mathcal{C}^{(u)}\big)$ and $p\big(y \mid \boldsymbol{x}, \mathcal{C}^{(l)}\big)$ by Gaussian Kernel Density Estimates (KDEs), where $|\mathcal{C}^{(u)}| + |\mathcal{C}^{(l)}| = N^{(u)} + N^{(l)} = |\mathcal{C}|$.

$\mathcal{C}^{(l)}$ and $\mathcal{C}^{(u)}$ are obtained by the same splitting algorithm that yields y^γ, to give the lower quantile and upper quantile sets.. The bandwidth of the kernels used in each density estimate is selected by a Scott's rule heuristic, and a clipping function to ensure a reasonable range within a uniform distribution. [14] In this study, 24 candidates from the default configuration of *TPE* were used as the number of initial candidates to initialise the BO iteration to construct $\mathcal{C}$ and finally obtain $\boldsymbol{x}$. (This is referred to as `n_startup_trials` in [14]) *Optuna*'s budget was constrained to 250 BO iterations, or, *Optuna* trials. For categorical hyper-parameters, *Optuna* selects them separately by maximising the ratio of density of those hyper-parameters appearing lower and upper quantile trials. That is, $\boldsymbol{x} = \arg\max_{\boldsymbol{x}} u(\boldsymbol{x})/l(\boldsymbol{x})$.

Note that, while tuning in ML involves to process of evaluating the models on a hold-out validation set per tuning trial, we utilise the *Optuna* framework implemented through the define-by-run style *Python* library [15], purely as an informed search algorithm on the model space by fitting models that differ by hyper-parameter tuning in each step. Due to the simplicity of re-fitting *LightGBM* compared Deep Learning models, pruning was not used in this application. It is also our aim to utilise such tuning frameworks that motivates our choice of *LightGBM*, due to its computational efficiency, yet effectiveness with continuous data, as detailed in earlier passages.

Post-hoc analyses with *Optuna* studies are also possible to assess the contribution of each hyper-parameter to the overall model performance. The following *LightGBM* hyper-parameters were tuned over the defined space in Table 7.

log[] specifies that the grid is log-spaced.[3] Those unspecified are linearly spaced. The learning rate $\eta = 0.05$ was set. The grid for `num_leaves` is dependent on `max_depth` as per the advice provided by [16], in which `num_leaves` $\leq 2^{\texttt{max\_depth}}$ was recommended. In this work however, we conservatively selected $2^{\texttt{max\_depth}-1}$ as the upper bound instead. Other unspecified hyper-parameters as found in [16] were set

[3]If a sequence $x_1, x_2, ..., x_n > 0$ are log-spaced, then $\dfrac{x_2}{x_1} = \dfrac{x_3}{x_2} = ... = \dfrac{x_n}{x_{n-1}} = \left(\dfrac{x_n}{x_1}\right)^{\frac{1}{n-1}}$

to default values. Hence, the tuning space defined in this research is limited to that defined in the table above to closely investigate the complexity of the best fit *LightGBM* on the panel datasets in terms of depth of boosting trees and number of leaves.

2.5 Explainable Artificial Intelligence

Often times, ML methods are seen as black box methods due to the density of parameters within their architectures. This poses a challenge to model interpretation. Hence, it has been a trend in research —especially those involving GBDT methods such as [13], [17], and [18]—to construct models that explain predictions at both a global and instance-specific level, an approach known as eXplainable Artificial Intelligence (xAI).

Because there is no one-size-fits-all method for xAI, we employ complementary techniques: Partial Dependence Plots (PDP) to assess global feature behavior, and Shapley values (SHAP) to isolate exact feature contributions to individual predictions. In the latter objective, we specifically investigate the beginning of Ramadan 2023 and 2024 that incited the large negative demand shifts highlighted in our EDA.

Specifically, we employ Partial Dependence Plots (PDP) to establish the baseline behavior of the LightGBM model. By marginalising over the distribution of all features in the complement set C, the PDP isolates the expected global effect of a target feature subset S:

$$\hat{f}_S(\boldsymbol{x}_S) = \frac{1}{n}\sum_{i=1}^{n} f\left(\boldsymbol{x}_S, \boldsymbol{x}_{i,C}\right) \tag{26}$$

This visualises whether relationships—such as the impact of applied discounts or rolling mean statistics—are linear or complex across the standard calendar year.

For instance-level interpretation, particularly when zooming in on specific timeframes like the Ramadan structural breaks, we utilise SHAP (SHapley Additive exPlanations). SHAP calculates the exact marginal contribution of a feature across all possible feature coalitions. For a feature j in the set of all features F, its Shapley value φ_j is given by:

$$\varphi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)![f(S \cup \{j\}) - f(S)]}{|F|!} \tag{27}$$

We favour SHAP over standard Global Feature Importance (GFI) or Permutation Feature Importance (PFI). GFI unfairly favours high-cardinality features, while PFI breaks the severe temporal correlations in our dataset. SHAP averts this by conditionally computing marginal expectations, ensuring that correlated features like `Revenue.lag1` and rolling means do not create impossible data states during evaluation.

Crucially, this interpretability framework serves as methodological triangulation for our econometric findings. Because our earlier Hsiao tests firmly rejected panel homogeneity, we specifically evaluated the models trained with panel categorical indicators to verify if the machine learning architecture autonomously captured the individual fixed effects.

Global Feature Contributions

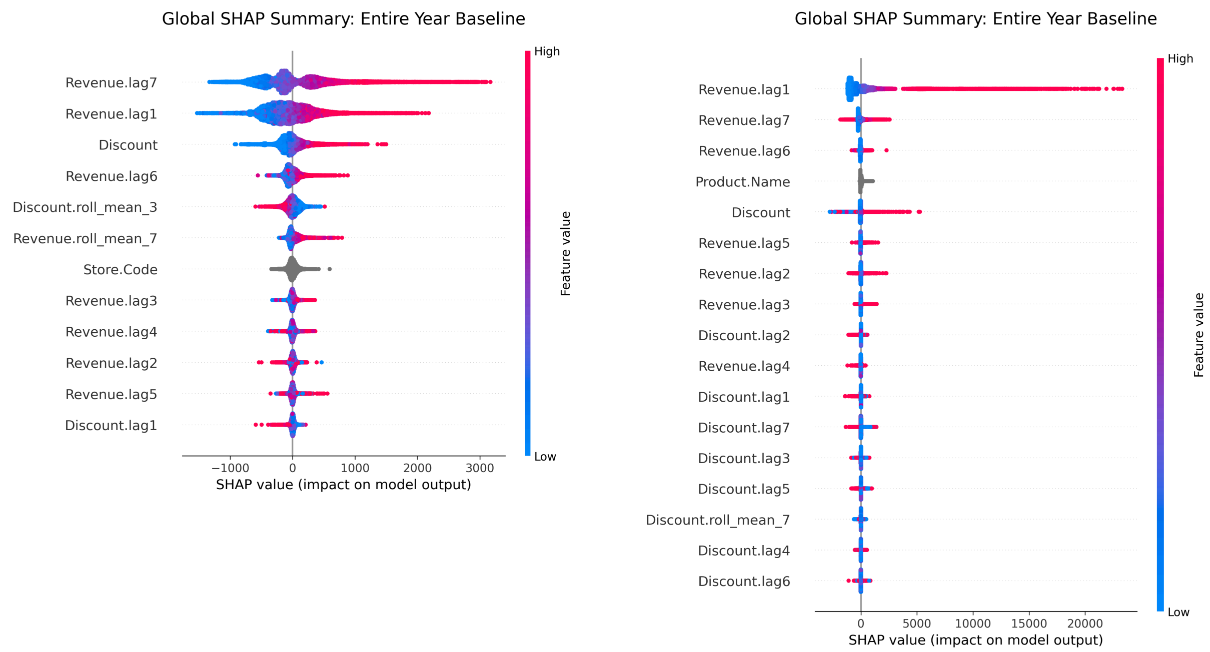


Figure 10: Global SHAP Summary plots for Store panels (left) and Drink panels (right).

As Figure 10 shows, feature importance differs significantly between the two panel types. For stores, *Revenue.lag7*, *Revenue.lag1*, and *Discount* are the main predictors, which supports our HNC test findings on weekly seasonality and promotional sensitivity. In contrast, drink-level revenue is largely driven by the previous day's performance (*Revenue.lag1*). Although *Discount* ranks 5th for drinks, its overall impact is much smaller compared to this first lag. This matches the sparse Granger causality we observed beyond the first day for individual products. Finally, the categorical indicators (*Store.Code* and *Product.Name*) rank 7th and 4th respectively, suggesting that while location and product type matter, the models rely more heavily on recent sales trends than static identifiers.

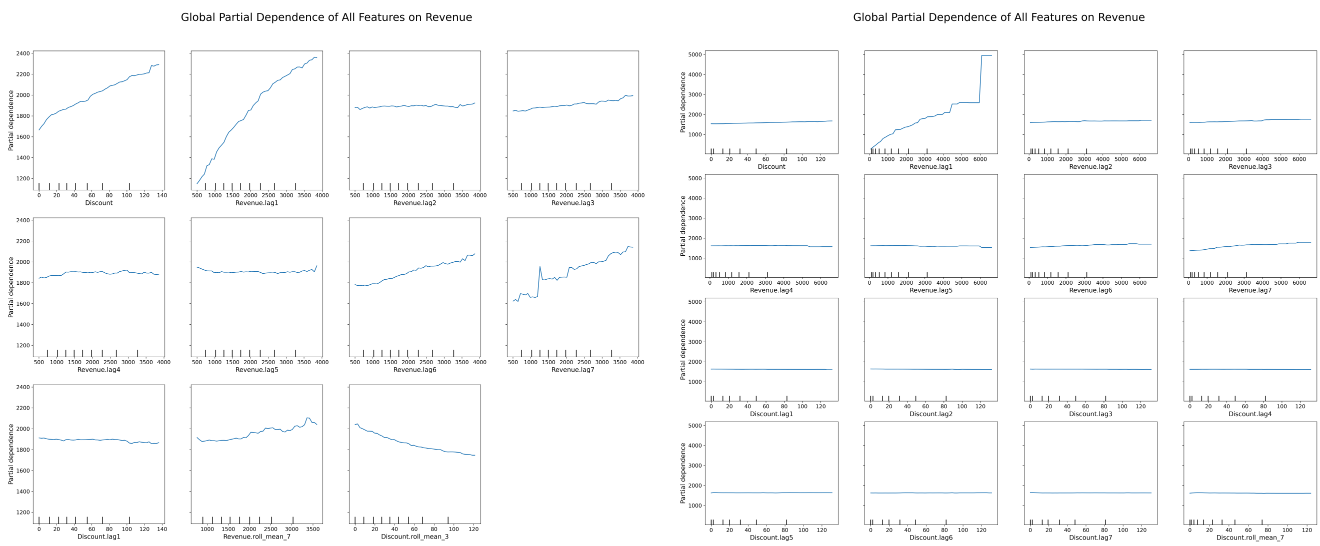


Figure 11: Global Partial Dependence Plots on Revenue for Store panels (left) and Drink panels (right).

The partial dependence plots in Figure 11 provide evidence of promotional cannibalisation. In the store model, the contemporaneous *Discount* PDP trends upward, while the *Discount.roll_mean_3* PDP slopes downward. This mirrors our CCEMG results (detailed later in Section 3.2): a strongly positive contemporaneous coefficient ($\beta^{(0)}_{\text{MG}}$) followed by significantly negative lags ($\beta^{(1)}_{\text{MG}}$, $\beta^{(2)}_{\text{MG}}$). Both frameworks independently confirm inter-temporal substitution, where consecutive discounts cannibalise future full-price sales. Drink-level PDPs for *Discount* remain flat, reinforcing that product-level demand is highly inelastic.

Localised Interpretability: Ramadan Structural Breaks

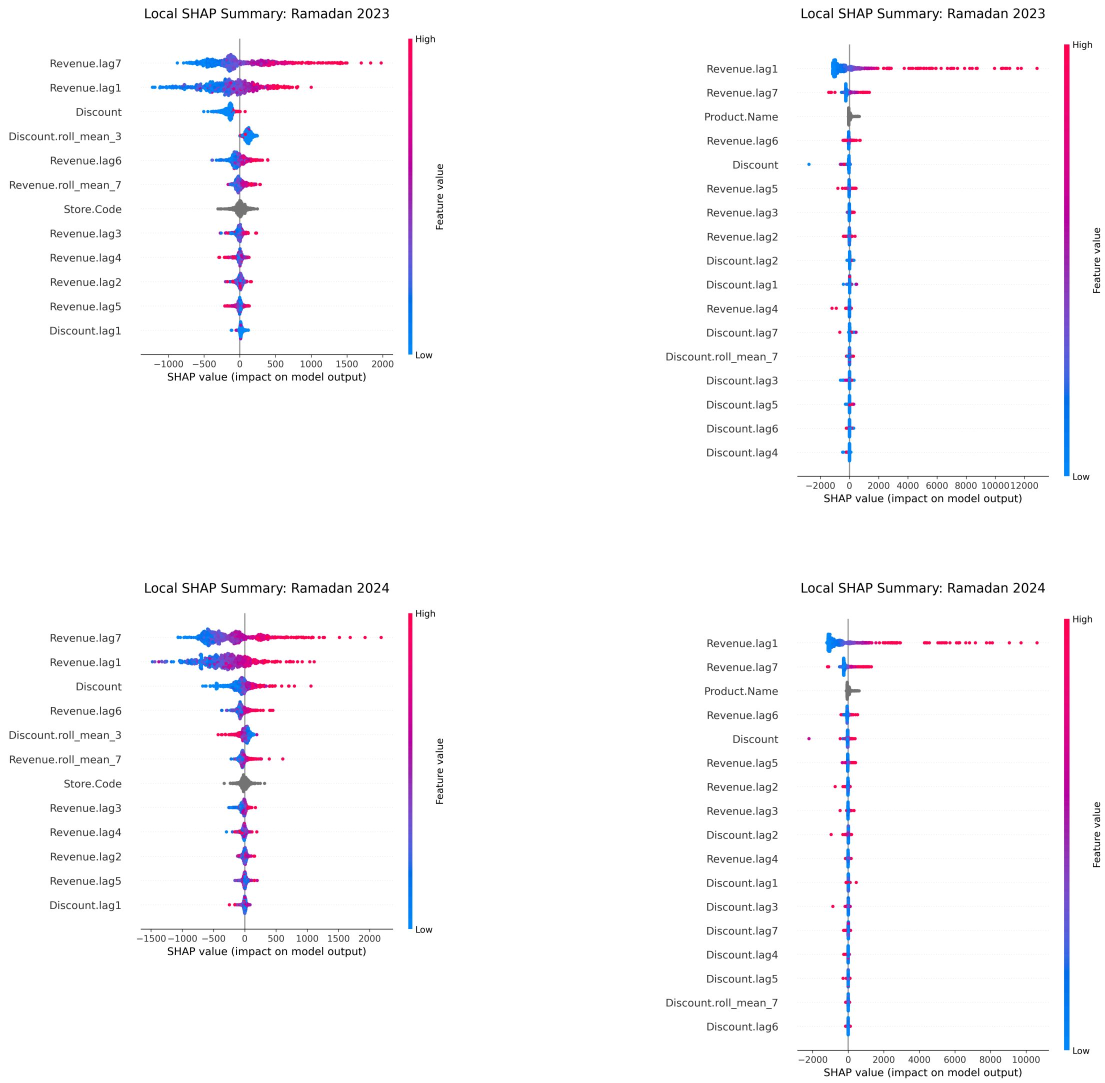


Figure 12: Local SHAP Summary plots for Ramadan structural breaks (2023 and 2024), Store panels (left) and Drink panels (right).

Localised summaries during Ramadan structural breaks reveal shifting importance during demand shocks. For stores, *Discount* remains a top driver, indicating that promotions are vital for maintaining foot traffic during the fasting month. For drinks, lag features entirely overshadow promotions, suggesting consumption is dictated by localized habit rather than price shifts.

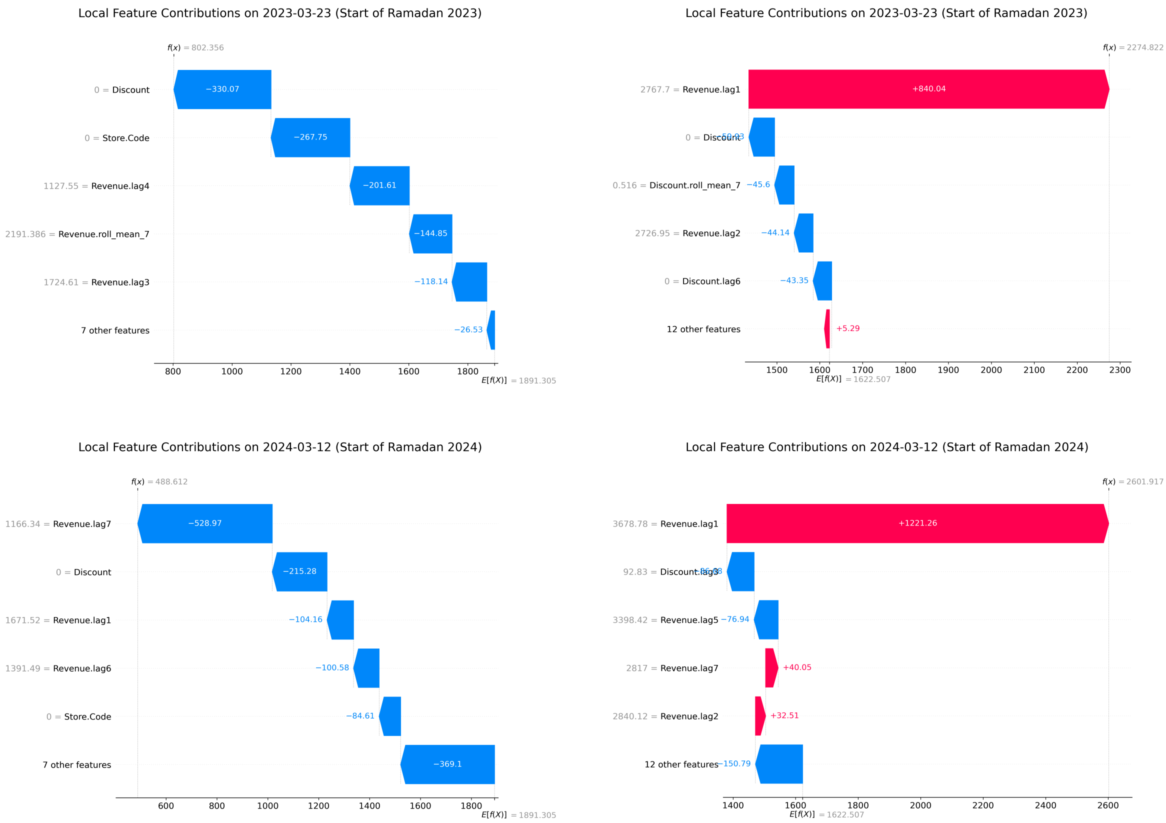


Figure 13: Local SHAP Waterfall plots for the onset of Ramadan in 2023 (top) and 2024 (bottom) for Store panels (left) and Drink panels (right).

Waterfall plots at the onset of Ramadan (Figure 13) quantify these penalties. In 2023, the absence of a promotion ($Discount = 0$) penalized store revenue by -330.07 units, whereas drinks saw a minor -50.93 unit penalty, buoyed instead by a $+840.04$ unit *Revenue.lag1* contribution. This pattern held in 2024, with stores incurring a -215.28 unit discount penalty while drinks received a massive $+1221.26$ unit boost from recent history (*Revenue.lag1*).

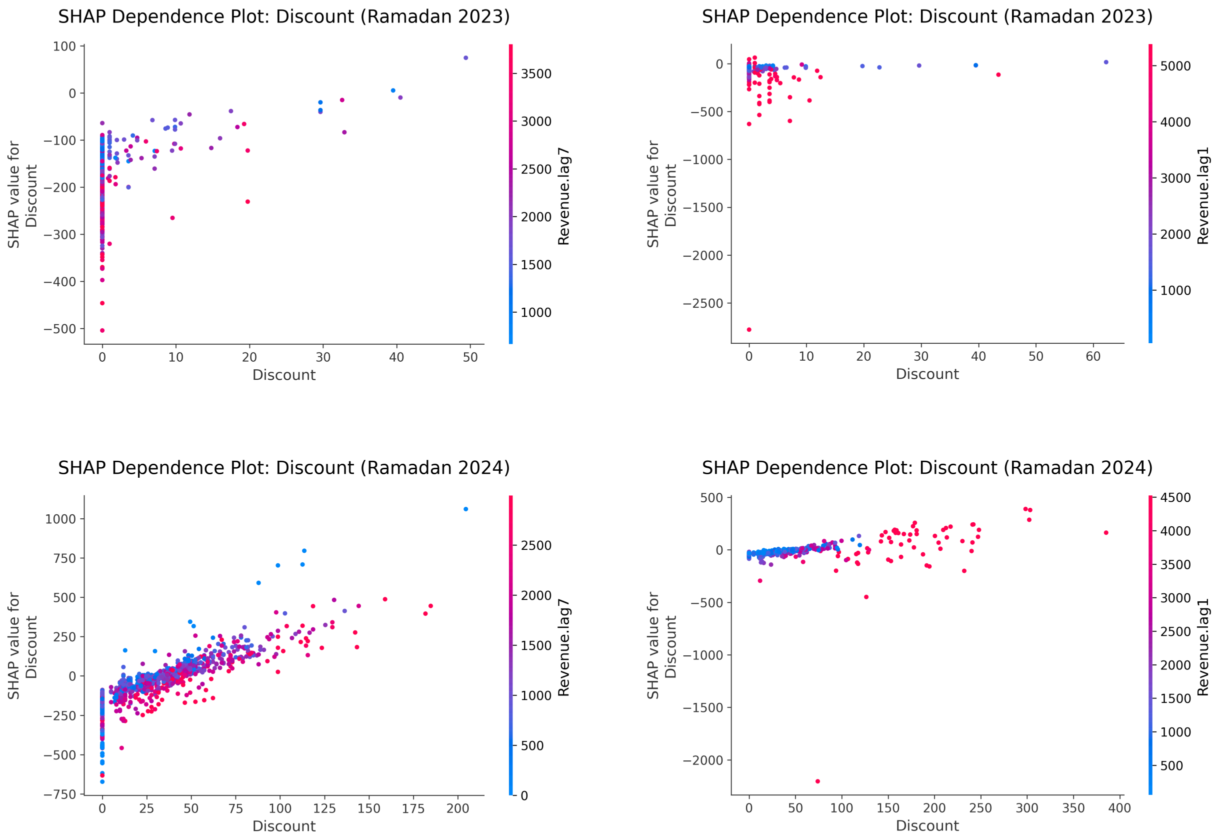


Figure 14: SHAP Dependence Plots for Discount during Ramadan periods (2023 and 2024) for Store panels (left) and Drink panels (right).

Finally, the SHAP dependence plots (Figure 14) show how promotions interact with other features during the fasting months. In the store models, the wide spread of points at $Discount = 0$ shows that offering no promotion generally leads to a large drop in predicted revenue. As the discount increases, the SHAP values go up. We colored these points by *Revenue.lag7* because it was the strongest time-based predictor for store traffic in our earlier global analysis. This color interaction highlights a key finding: the biggest benefits from large discounts go to store-days that had low historical weekly traffic (the blue points). This points to diminishing returns for stores that are already busy. In practice, heavy store-wide discounts act mostly as a way to boost underperforming stores, rather than increasing sales at locations that already have high foot traffic. For the drink-level models, sales are driven heavily by momentum from the previous day (*Revenue.lag1*), which our global analysis identified as the most important feature. While the 2024 plot shows a slight upward trend for very high discounts, the interaction shows that the few drinks that actually got a big boost from these discounts were the ones that already had high sales the day before (red points). Applying large discounts to drinks with low recent sales (blue points) results in a flat, near-zero SHAP value. This highlights that product-level demand is inelastic and largely driven by consumer habits. While price cuts might slightly boost an already-popular drink, they do not create new demand for items that customers rarely buy.

3 Results and Discussion

3.1 Causality Analysis and Feature Selection

To evaluate the true causal dynamics across the dataset and justify our model specifications, we visualised the individual Granger causality p-values across all lag orders via bar plots (see Appendices H and I), explicitly comparing the results before and after applying the CCEMG projection. In the uncorrected models used in the HNC tests, the data presented an illusion of widespread, highly significant Granger causality. Because the panels share massive unobserved common shocks—such as nationwide promotional campaigns, holidays, or broad macroeconomic shifts—the uncorrected models suffered from severe cross-sectional dependence (as evidenced by the high uncorrected $\overline{\rho}$ and $\tilde{Z}$ statistics in Table 5 and Table 6).

The standard test incorrectly attributed the variance from these shared, systemic shocks directly to the lagged discounts, leading to inflated Wald statistics and deceptively low p-values across almost all lags and panels. Stores such as 0-24 that were open throughout the years 2023-2024 had significant evidence against Granger Non-causality, which we may tie to a promotional event that likely took place after Ramadan ‘23. During this period, discount depth totalled across all drinks peaked between June and July 2023, as seen in Figure 3.

However, once the CCEMG projection was applied to purge these unobserved common factors, the true underlying relationships emerged. The corrected plots reveal significant cross-sectional heterogeneity, indicating that discounts do not uniformly drive revenue. Much of the previously observed test statistics in Granger causality was artificially inflated by global effects; upon controlling for these common shocks, it becomes evident that each panel reacts to the discount strategies they are exposed to differently, but mostly do not show result in Granger causation.

For the store-level panels, the corrected data shows that several locations (e.g., Stores 0, 1, and 8) still exhibit significant evidence of causality at early lags (Lags 1–3), which typically fades by Lags 6 and 7. At the store level, these early lags successfully capture the cumulative carryover effect of promotional footfall throughout the week. Because this creates a strong, structured trend across the store panels, it empirically justifies the inclusion of all 7 historical discount lags in our store-level models to fully capture the promotional cycle and prevent omitted-variable bias.

In contrast, the corrected Granger causality of the discount depth at the drink-level is sparse, with most products failing to cross the significance threshold beyond Lag 1. Product-level sales are highly volatile and heavily skewed by daily factors like consumer routine or localised weather, which easily mask the direct impact of a product-specific promotion. Because granular product pricing effects are erratic and non-linear, incorporating extensive historical discount lags would only introduce noise and cause over-parameterisation.

3.2 CCEMG

The large magnitudes of panel-pairwise correlation coefficients motivate the application of CCEMG in this work by projecting the data to purposefully purge the common factor term present in the residuals of VCM. CCEMG augments the model with a projection on cross-sectional averages followed by subtracting from original data to purge the influence of unobserved common factors. The heat maps in the appendices visualise the before and after effects of this projection on the pairwise correlation coefficient magnitudes.

Table 8: Panel (Store) MG Coefficient Estimation

Coef.	Est.	SE	z-val	p
$\gamma^{(1)}_{\mathrm{MG}}$	0.2721	0.0211	12.8667	2.2×10^{-16}
$\gamma^{(2)}_{\mathrm{MG}}$	0.0573	0.0117	4.8973	9.72×10^{-7}
$\gamma^{(3)}_{\mathrm{MG}}$	0.0356	0.0085	4.1733	3.00×10^{-5}
$\gamma^{(4)}_{\mathrm{MG}}$	0.0280	0.0093	3.0138	0.0026
$\gamma^{(5)}_{\mathrm{MG}}$	-0.0076	0.0138	-0.5503	**0.5821**
$\gamma^{(6)}_{\mathrm{MG}}$	0.0414	0.0087	4.7496	2.04×10^{-6}
$\gamma^{(7)}_{\mathrm{MG}}$	0.1457	0.0216	6.7482	1.50×10^{-11}
$\beta^{(0)}_{\mathrm{MG}}$	3.4492	0.3421	10.0829	2.2×10^{-16}
$\beta^{(1)}_{\mathrm{MG}}$	-0.4196	0.1287	-3.2614	5.24×10^{-4}
$\beta^{(2)}_{\mathrm{MG}}$	-0.3396	0.1120	-3.0310	4.53×10^{-3}
$\beta^{(3)}_{\mathrm{MG}}$	-0.1473	0.0952	-1.5471	**0.1218**
$\beta^{(4)}_{\mathrm{MG}}$	-0.2564	0.1255	-2.0425	0.0411
$\beta^{(5)}_{\mathrm{MG}}$	0.0589	0.1480	0.3978	**0.6908**
$\beta^{(6)}_{\mathrm{MG}}$	-0.2718	0.1129	-2.4073	3.86×10^{-3}
$\beta^{(7)}_{\mathrm{MG}}$	-0.1416	0.1344	-1.0530	**0.2923**
R^2				0.8210
$\overline{\rho}$				-0.022231
$\lvert\overline{\rho}\rvert$				0.1133

Table 9: Panel (Drink) MG Coefficient Estimation

Coef.	Est.	SE	z-val	p
$\gamma^{(1)}_{\mathrm{MG}}$	0.2573	0.0206	12.5149	2.2×10^{-16}
$\gamma^{(2)}_{\mathrm{MG}}$	0.1254	0.0084	14.9729	2.2×10^{-16}
$\gamma^{(3)}_{\mathrm{MG}}$	0.0583	0.0081	7.1676	7.63×10^{-13}
$\gamma^{(4)}_{\mathrm{MG}}$	0.0235	0.0090	2.6115	9.02×10^{-3}
$\gamma^{(5)}_{\mathrm{MG}}$	0.0218	0.0068	3.2173	1.29×10^{-3}
$\gamma^{(6)}_{\mathrm{MG}}$	0.0309	0.0083	3.7284	1.93×10^{-4}
$\gamma^{(7)}_{\mathrm{MG}}$	0.0856	0.0081	10.5862	2.2×10^{-16}
$\beta^{(0)}_{\mathrm{MG}}$	0.6978	0.0813	8.5798	2.2×10^{-16}
$\beta^{(1)}_{\mathrm{MG}}$	0.0532	0.0695	0.7645	**0.4445**
$\beta^{(2)}_{\mathrm{MG}}$	-0.0195	0.0666	-0.2920	**0.7703**
$\beta^{(3)}_{\mathrm{MG}}$	-0.0048	0.0604	-0.0794	**0.9367**
$\beta^{(4)}_{\mathrm{MG}}$	0.0005	0.0586	0.0078	**0.9938**
$\beta^{(5)}_{\mathrm{MG}}$	-0.1060	0.0533	-1.9895	0.0466
$\beta^{(6)}_{\mathrm{MG}}$	-0.0050	0.0712	-0.0700	**0.9442**
$\beta^{(7)}_{\mathrm{MG}}$	0.0444	0.0590	0.7522	**0.4520**
R^2				0.9436

Table 10: Final Panel (Drink) MG

Coef.	Est.	SE	z-val	p
$\gamma^{(1)}_{\mathrm{MG}}$	0.2635	0.0206	12.7857	2.2×10^{-16}
$\gamma^{(2)}_{\mathrm{MG}}$	0.1263	0.0084	15.0422	2.2×10^{-16}
$\gamma^{(3)}_{\mathrm{MG}}$	0.0604	0.0083	7.2879	3.15×10^{-13}
$\gamma^{(4)}_{\mathrm{MG}}$	0.0270	0.0092	2.9303	3.39×10^{-3}
$\gamma^{(5)}_{\mathrm{MG}}$	0.0234	0.0069	3.3811	7.22×10^{-4}
$\gamma^{(6)}_{\mathrm{MG}}$	0.0339	0.0083	4.0760	4.58×10^{-5}
$\gamma^{(7)}_{\mathrm{MG}}$	0.0913	0.0078	11.6472	2.2×10^{-16}
$\beta^{(0)}_{\mathrm{MG}}$	0.6209	0.0646	9.6170	2.2×10^{-16}
R^2				0.9420
$\overline{\rho}$				-0.0010619
$\lvert\overline{\rho}\rvert$				0.059378

This projection is imperative in constructing the models, and not just running HNC, because it may give misleading individual p-values on the significance of the covariate coefficients in modelling the panel-wise revenue. Under the VCM framework in Equation 2, coefficients are likely significant in explaining the target variable through the common factor term as a proxy. Moreover, coefficients are also biased in order to account for modelling the common factor term if they are not corrected out of the model. As we have seen that the common factors each panel's regressors has some contribution to the common factor in terms of cross-sectional sample means, unpurged common factors in the residuals give rise to endogeneity, where regressors are correlated to the residuals through the common factors. [19] discusses endogeneity bias affecting the estimates of coefficients.

Therefore, to ensure this form of bias is minimal in the estimation of the coefficients, we will assume that purging the common factors, and hence cross-sectional dependence, by a projection to obtain Equation 18 is sufficient. Serial correlation within the residuals of the final models are observed by inspecting individual ACF plots. The results presented in the above tables correspond to the MG coefficient estimates obtained from the CCEMG model. Given the large number of panels in the dataset, statistical significance is assessed at the level of the MG estimates rather than at the level of individual panel-specific coefficients. On all the models presented above in Table 8, Table 9, and Table 10, we ran the Maddala-Wu diagnostic tests which informed us of residual stationarity with $m = 0$.

Whilst fitting the models without a contemporaneous discount term included, the insignificance of lagged terms did align with the HNC test results under a common correlated effects correction. Although the null of HNC was rejected after computing the $\tilde{Z}$ statistic from $\overline{W}_N$, inspecting individual bar plots of the p-values of HNC allows us to note that across all lags, Granger causation was quite sparse across stores, and a small p-value was only attributed to a few stores. Inspecting the significance of each variable through the MG estimators showed that these lags were insignificant.

However, including the contemporaneous variable for discount for both series led to significant lagged variables of discount in both the store- and drink- level models, and significance of the contemporaneous variable itself. Hence, this lead us to interpret that the current conscious decision for the depth of discount in the sales made within each store, or on each beverage is a statistically significant variable in modelling the revenue generated by the i-th store/drink. Therefore, based on these results and supporting literature [20], [21], we will include the contemporaneous discount series as predictors in models of both levels.

Store-level effects

There are some interesting interpretations in the coefficients of the discount series. Firstly, more recent historical discount terms were more significant in the model than the longer-term lags. Moreover, lag terms of $\beta^{(q)}_{\text{MG}}$, where $q = 3, 4$ appear to have shrunk compared to smaller lag coefficients where $q = 1, 2$. Additionally, for both revenue and discount, some mid-length lags were insignificant. This is attributed to the multi-collinearity between the lags of revenue; revenue term is highly persistent, and with information from the start and end of a 7-lag historical window available, the variable that carries information from the middle of the window are not significant anymore.

Moreover, the coefficients of CCEMG on the store-level suggest that, the current depth of discount has an average positive impact of approximately 3.45% the current day's revenue. However, this causes an average negative -0.42% impact on the revenue generated in the next day. These align with the literature [21] — the depth of discount tied to sales not creating entirely new demand, but is partly borrowing from the immediate future.

Drink-level effects

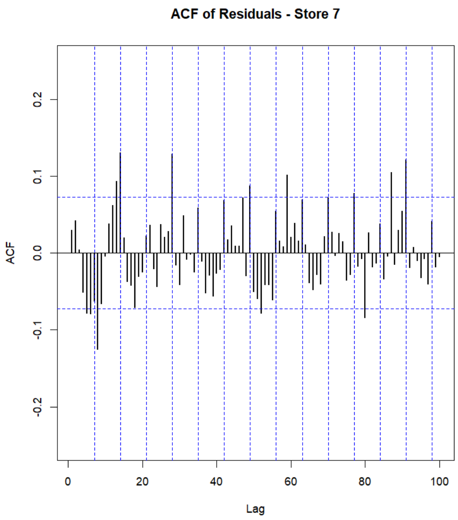
The estimated mean-group coefficients for the drink-level panel reveal some clear patterns. Recent revenue lags are highly significant, particularly the first, second, and third lags, indicating that short-term historical revenue is a strong predictor of current revenue. Longer lags remain positive but smaller, reflecting some decay in predictive power over time. For the discount series, most lag coefficients are statistically insignificant, except for the fifth lag, which shows a modest negative effect. This suggests that, at the drink level, current revenue is primarily driven by recent revenue history, while discounting effects are generally weak or delayed. Hence, we were able to fit a more parsimonious drink-level panel CCEMG, as presented in Table 10.

Overall, the results indicate that short-term persistence dominates revenue dynamics at the drink level, and discount interventions have limited immediate impact. The only noticeable effect is a slight negative influence from the fifth lag of discount, implying a delayed or weaker carry-over effect compared to the store-level panel.

3.3 LightGBM and a hybrid-stack LightGBM

In modelling the revenue by a full-stack *LightGBM*, we have the option to include a category-factored panel identifier variable in the input matrix of *LightGBM* to allow *LightGBM* to acknowledge the panel nature of the dataset. Below are the best architectures of *LightGBM* to fit to the data defined by the set of *Optuna* fine-tuned hyper-parameters.

In another modelling approach presented in this work, we are also able to develop a hybrid-stack framework, in which the residuals of each of the CCEMG models fitted on each panel-aggregated dataset were extracted, then fed into a *LightGBM* architecture. A hybrid model setup is inspired in this study due to observations in the previously fitted CCEMG models. Firstly, while structural breaks have already been acknowledged, the model does not at all include any variables that account for them. Secondly, some models at the store- and drink-level poorly captured the seasonality in the data. This was observable in the individual ACF plots for particular store/product.



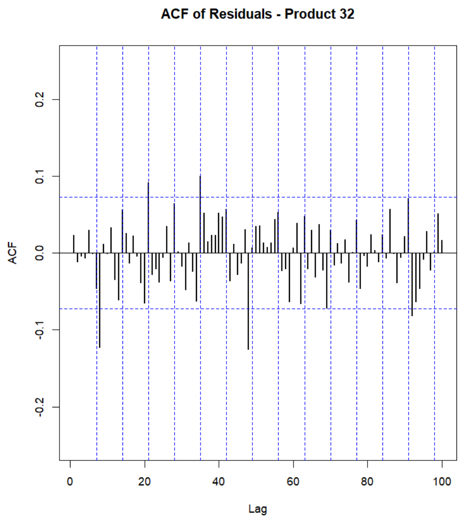


Table 11: LightGBM hyper-parameters for each model and target.

Parameter	Store		Drink	
	Full-Stack	**Hybrid**	**Full-Stack**	**Hybrid**
feature_fraction	0.9899	0.6156	0.9899	0.9758
max_depth	16	16	16	16
num_leaves	17887	19707	32761	28535
min_child_samples	10	10	10	10
num_tree	250	250	250	250
lambda_l2	3.3×10^{-4}	5.3×10^{-5}	1.0×10^{-8}	3.8×10^{-8}

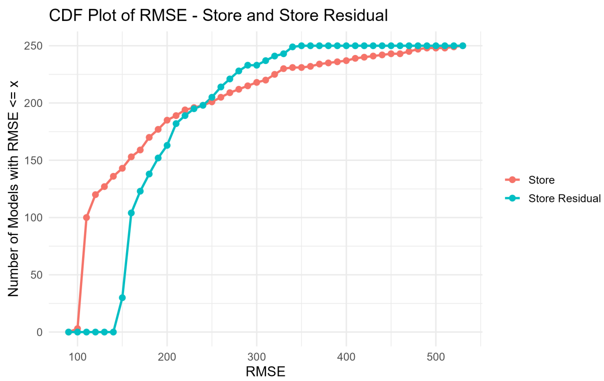


Figure 15: CDF of RMSE of Full-Stack and Hybrid Models (Store)

Table 12: Feature engineering parameters for each model and target.

Parameter	Store		Drink	
	Full-Stack	Hybrid	Full-Stack	Hybrid
K	5	6	7	7
Q	7	1	5	2
W_r	7	7	7	(3, 7)
W_d	0	7	7	0
R^2	**0.9946**	0.9379	**0.9988**	0.9548

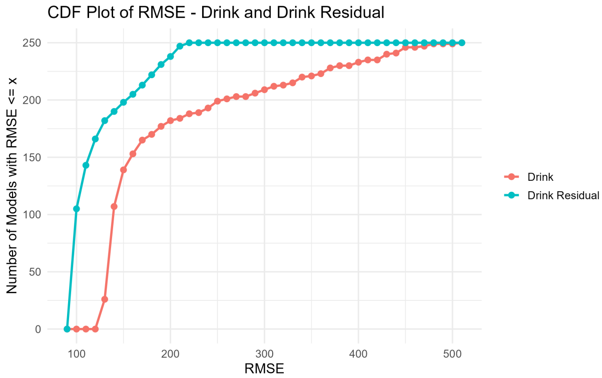


Figure 16: CDF of RMSE of Full-Stack and Hybrid Models (Drink)

The full-stack and hybrid models do not differ much in terms of the magnitudes of the hyper-parameters.

The plots above present the cumulative distribution functions (CDFs) of the RMSE objective values obtained from 250 models evaluated over 250 Optuna trials for each specification. Based on these trials, the hyper-parameter configurations required to achieve strong predictive performance in full-stack LightGBM for store-level models do not appear to be highly unique. This is reflected in the steep initial rise of the RMSE CDF, where approximately 100 configurations explored during BO attained RMSE values of ≤ 100. This suggests that accurate store-level full-stack LightGBM models can be obtained under a relatively broad range of hyper-parameter settings.

In contrast, the hybrid-stack LightGBM store models appear to be more sensitive to hyper-parameter selection. Only a relatively small number of configurations (approximately 25) yielded comparably accurate predictive performance, indicating that good-performing hybrid-stack specifications are concentrated within a narrower region of the hyper-parameter space.

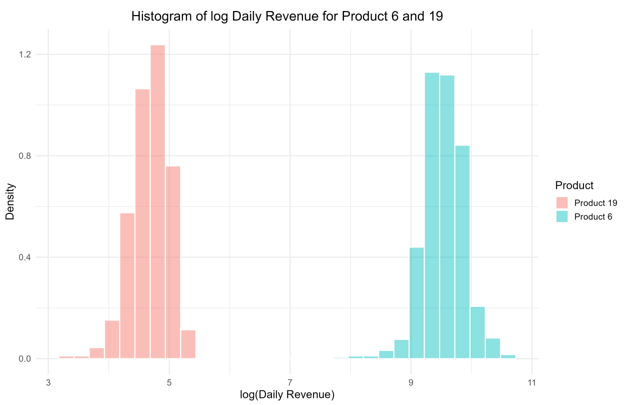


Figure 17: Histogram of the realised $\log(y_{i,t})$ for $i = 6, 19$.

The opposite pattern is observed for drink-level models. However, in the evaluation stage of *LightGBM* fitted on the drink-panel aggregated dataset, standardisation must be applied to each panel on the target variable (revenue), and subsequently its lags, and the discount series as well. That is, subtracting and then dividing by panel estimated mean and standard deviations. This step is necessary, since upon inspecting the within sample forecasts, the *LightGBM* models on the drink data was not able to capture vastly varying scales of time series variation between each panels. The histogram below demonstrate this large difference in the distribution of the realisations of the DGP of two panels.

For these, the hybrid-stack *LightGBM* exhibited greater robustness to hyper-parameter selection, with approximately 100 configurations yielding accurate predictive performance. Meanwhile, only around 25 configurations of the full-stack *LightGBM* achieved similarly low RMSE values, suggesting that accurate drink-level full-stack models depend on a more restricted set of hyper-parameter combinations.

While both full-stack and hybrid models yielded excellent R^2 scores, full-stack models on both panel datasets had exceedingly high R^2 scores. The residuals within the hybrid models were generally more evenly distributed with little to no outliers in each panel's forecast, unlike the full-stack models, where the errors were generally smaller with a few cases of structural breaks that were not captured well by *LightGBM*. The trade-off between performance is not significant at all for model interpretability and the hybrid stack *LightGBM* has proven that it (a) does not fit to noise from within the residuals generated by the first stage CCEMG, (b) detects structural breaks well. ACF plots of individual panels' residuals from both stacks showed that autocorrelation lied within their respective confidence bands.

4 Conclusion

This research highlights the necessity of controlling for unobserved heterogeneity and cross-sectional dependence when evaluating promotional effectiveness in retail networks. By applying a Common Correlated Effects Mean Group (CCEMG) estimator, this study effectively purged the systemic macroeconomic shocks that initially inflated standard panel Granger causality tests. The corrected models reveal a distinct contrast in revenue dynamics between store-level and product-level aggregations.

At the store level, econometric outputs and global Partial Dependence Plots (PDP) confirm the presence of inter-temporal substitution, where contemporaneous discounting generates immediate revenue spikes but cannibalises subsequent sales. However, localised SHAP evaluations during structural demand shocks (such as the onset of Ramadan) indicate that aggressive store-wide discounting acts as a crucial compensatory strategy. The highest marginal benefits of these discounts are concentrated in store-days with low historical footfall, effectively stabilising revenue during underperforming periods.

Conversely, both statistical and machine learning frameworks indicate that product-level demand is highly inelastic. Drink revenue is primarily dictated by a momentum effect driven by prior-day sales, strongly suggesting that individual beverage consumption is anchored in daily routines and ingrained taste preferences rather than immediate price sensitivity. Consequently, while substantial price reductions provide a marginal lift to already-popular beverages, they consistently fail to disrupt established consumer habits or generate new demand for infrequently purchased items.

Methodologically, this study demonstrates the viability of a hybrid forecasting architecture. By utilising rigorous econometric projections to account for structural breaks and common factors, the subsequent *LightGBM* models achieved high predictive accuracy and interpretability while avoiding over-parameterisation and noise-fitting. Operationally, these findings suggest that retail pricing strategies should pivot away from granular, product-specific price reductions and instead optimise store-wide promotional depth to selectively target and stimulate low-traffic periods.

The models developed demonstrate the tradeoff between predictive power and interpretability. By implementing a hybrid-stack framework, we yield the benefits of the interpretability of the first stage CCEMG, followed by a *LightGBM* stack which captures the seasonality and structural breaks. It also serves as a good sign, since we do not wish for the model frameworks to achieve an exceedingly high R^2 by capturing noise in the data.

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5 Appendices

Appendix A : One-one map for store code and product name to integer encoding

Table 13: Store Code Index Mapping

Index	Store Code	Index	Store Code	Index	Store Code
0	S0003	14	S0029	28	S006
1	S0007	15	S0034	29	S0062
2	S0009	16	S0037	30	S0065
3	S001	17	S0039	31	S0080
4	S0010	18	S0040	32	S0083
5	S0011	19	S0043	33	S0084
6	S0012	20	S0044	34	S0091
7	S0015	21	S0045	35	S0092
8	S0016	22	S0047	36	S0097
9	S0018	23	S0049	37	S0099
10	S0019	24	S0050	38	S0105
11	S0020	25	S0054	39	S0110
12	S0024	26	S0057		
13	S0026	27	S0058		

Table 14: Drink Index Mapping

Index	Product Name	Index	Product Name	Index	Product Name
0	Americano	13	Dark Chocolate Latte	26	Matcha Latte
1	Caffe Latte	14	Dark Chocolate Mocha	27	Matcha Frappe
2	Cappuccino	15	Double Cocoa Mocha Latte	28	Mocha
3	Caramel Cream Frappe	16	Double Mocha Frappe	29	Nutty Almond Latte
4	Caramel Espresso Frappe	17	English Breakfast Tea	30	Oolong Tea
5	Chamomile Tea	18	Espresso Frappe	31	Pecan Crunch Latte
6	Cheesecream Latte	19	Espresso	32	Rose Latte
7	Chocochip Frappe	20	Hazelnut Latte	33	Salted Caramel Macchiato
8	Citrus Blue Frappe	21	Lemon Lime Fizz	34	Salted Choco Cream Mocha
9	Classic Caramel Macchiato	22	Lemon Tea Fizz	35	Special Cookies Latte
10	Cocoa Latte	23	Mango Lime Fizz	36	Vanilla Latte
11	Cookies Cream Frappe	24	Mango Lime Frappe		
12	Dark Chocolate Frappe	25	Mango Strawberry Fizz		

Appendix B : Table of Notations

Symbol	Description
i	Cross-sectional unit, $i = 1, ..., N$
K, Q	Maximum lag length
N	Number of cross-sectional units
T_i	Number of observations for the i-th panel
$y_{i,t}$	Dependent variable for the i-th panel at time t
$x_{i,t}$	Explanatory variable for unit i at time t
α_i	Fixed (Individual) effect of the i-th panel
$x_{i,t}^{(k)}$	values of the of the regressor x lagged by k for the i-th panel, at time t
$\boldsymbol{x}_i^{(k)}$	$(x_{i,1}^{(k)}, \cdots, x_{i,T_i}^{(k)})$
$\hat{\varepsilon}_{i,t}$	$y_{i,t} - \hat{y}_{i,t}$
$\hat{\boldsymbol{\varepsilon}}_i$	$\left[\varepsilon_{i,1}, ..., \varepsilon_{i,T_i}\right]$
$\tilde{\boldsymbol{\varepsilon}}_i$	$\hat{\boldsymbol{\varepsilon}}_i / \sigma_{\varepsilon,i}$
I_T	$T \times T$ identity matrix
B	Backshift Operator
WN	Gaussian White Noise

Appendix C : Lemmas and Theorems in Econometrics

Lemma 5.0.1 (First and Second moments of Z_i^2)

Let $Z_i \sim \mathcal{N}(0,1)$. Then, $\mathbb{E}(Z_i) = 0$, $\mathbb{E}(Z_i^2) = 1$.
Proof:
Both are clear from $Z_i \sim \mathcal{N}(0,1)$.
$\mathbb{E}(Z_i) = 0$, and $\mathbb{E}(Z_i^2) = \mathbb{V}ar(Z_i) + \mathbb{E}(Z_i)^2 = 1$. ■

♡

Theorem 5.0.2 (Asymptotic distribution of $\frac{\chi^2_{n-k}}{n-k}$)

Let $X \sim \chi^2_{n-k}$. Then, $\frac{X}{n-k} \xrightarrow{p} 1$.
Proof:
Since $X \sim \chi^2_{n-k}$, then, for some $Z_1, ..., Z_{n-k} \sim iid\mathcal{N}(0,1)$, $X = \sum_{i=1}^{n-k} Z_i^2$. As $\mathbb{E}(Z_i^2) = 1$ by Lemma 5.0.1, $\mathbb{E}(X) = n-k$. Thus, by the Weak Law of Large Numbers (WLLN), the sample mean variable $\overline{X} = \frac{1}{n-k}\sum_{i=1}^{n-k} Z_i^2 \xrightarrow{p} 1$ ■

♡

Theorem 5.0.3 (Slutsky's Theorem)

Let X_n be a sequence of random variables such that $X_n \xrightarrow{d} X$, and let Y_n be a sequence of random variables such that $Y_n \xrightarrow{p} c$, where c is a non-zero constant. Then,

$$\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c} \tag{28}$$

Proof:
The proof is omitted for brevity. For a formal mathematical proof, refer to standard probability and econometric literature [22].

♡

Corollary 5.0.3.1 (Distribution of $J \times F$)

Let F be as defined in Equation 8. Then, $J \times F \xrightarrow{d} \chi^2_J$
Proof:
$J \times F = \frac{\chi^2_J}{\chi^2_{n-k}/(n-k)}$. By Slutsky's Theorem, whereby, if $Y \xrightarrow{p} a \neq 0$, then, $X/Y \xrightarrow{d} X/a$.
Hence, by Theorem 5.0.2, $J \times F \xrightarrow{d} \chi^2_J$.

♡

Appendix D : Derivation of Wald test statistics distributions

Consider the individual Wald test statistic:

$$\begin{aligned}
W_i &= \hat{\boldsymbol{\theta}}_i^\top \boldsymbol{R}^\top \left(\sigma_i^2 \boldsymbol{R}(\boldsymbol{Z}_i^\top \boldsymbol{Z}_i)^{-1} \boldsymbol{R}^T\right)^{-1} \boldsymbol{R}\hat{\boldsymbol{\theta}}_i \\
&= \frac{\hat{\boldsymbol{\theta}}_i^\top \boldsymbol{R}^\top \left(\boldsymbol{R}(\boldsymbol{Z}_i^\top \boldsymbol{Z}_i)^{-1} \boldsymbol{R}^T\right)^{-1} \boldsymbol{R}\hat{\boldsymbol{\theta}}_i}{(T_i - 2K - 1)^{-1} \hat{\varepsilon}_i^\top \hat{\varepsilon}_i} \\
&= \frac{\tilde{\varepsilon}_i^\top \Phi_i \tilde{\varepsilon}_i}{(T_i - 2K - 1)^{-1} \tilde{\varepsilon}_i^\top M_i \tilde{\varepsilon}_i}, \\
&\quad \forall i = 1, \cdots, N \text{ panels, where} \\
&\quad \tilde{\varepsilon}_i = \frac{\varepsilon_i}{\sigma_{\varepsilon,i}}
\end{aligned} \tag{29}$$

and the average Wald test statistic:

$$\overline{W}^{\text{HNC}} = \frac{1}{N}\sum_{i=1}^{N} W_i \tag{30}$$

$$\begin{aligned}
\boldsymbol{\Phi}_i &= \boldsymbol{H}^T \left[\boldsymbol{R}(\boldsymbol{Z}_i^\top \boldsymbol{Z}_i)^{-1} \boldsymbol{R}^\top\right]^{-1} \boldsymbol{H} \\
\boldsymbol{H} &= \boldsymbol{R}(\boldsymbol{Z}_i^\top \boldsymbol{Z}_i)^{-1} \boldsymbol{Z}_i^\top
\end{aligned} \tag{31}$$

When estimating the error terms ε_i, $2K+1$ degrees of freedom are lost as a consequence of estimating $2K$ parameters and individual intercepts. By **A1**, $(T_i - 2K - 1)^{-1} \hat{\varepsilon}_i^\top \hat{\varepsilon}_i$ is $\chi^2(T_i - 2K - 1)$ distributed, and is a sample estimate of $\sigma^2_{\varepsilon,i}$. By asymptotic properties on T_i, $(T_i - 2K - 1)^{-1} \hat{\varepsilon}_i^\top \hat{\varepsilon}_i \xrightarrow{p} \sigma^2_{\varepsilon,i}$. Equivalently:

$$(T_i - 2K - 1)^{-1} \tilde{\varepsilon}_i^\top \boldsymbol{M}_i \tilde{\varepsilon}_i \underset{T_i \to \infty}{\xrightarrow{p}} 1 \tag{32}$$

$\because$,

$$\tilde{\varepsilon}_i^\top \boldsymbol{M}_i \tilde{\varepsilon}_i = \frac{1}{\sigma^2_{\varepsilon,i}} (\boldsymbol{M}_i \hat{\varepsilon}_i)^\top (\boldsymbol{M}_i \hat{\varepsilon}_i) = \frac{1}{\sigma^2_{\boldsymbol{\varepsilon},i}} \hat{\varepsilon}_i^\top \hat{\varepsilon}_i \tag{33}$$

Using Slutsky's theorem,

$$\widehat{W}_i = \frac{\tilde{\varepsilon}_i^\top \boldsymbol{\Phi}_i \tilde{\varepsilon}_i}{(T_i - 2K - 1)^{-1} \tilde{\varepsilon}_i^\top M_i \tilde{\varepsilon}_i} \xrightarrow{p} \tilde{\varepsilon}_i^\top \boldsymbol{\Phi}_i \tilde{\varepsilon}_i \tag{34}$$

The equality in Equation 33 is derived from the fact that $\boldsymbol{\varepsilon}_i = \boldsymbol{y}_i - \boldsymbol{Z}_i \boldsymbol{\theta}_i \implies \boldsymbol{M}_i \varepsilon_i = \hat{\varepsilon}_i$, $\because \boldsymbol{M}_i \boldsymbol{Z}_i = \mathbf{0}$.

Appendix E : Common Correlated Effects Mean Group model derivation

Given the panel VCM with a common factor term embedded in its residual $\boldsymbol{\varepsilon}_i$, we have the model:

$$\begin{aligned}
\boldsymbol{y}_i &= \alpha_i \mathbf{1}_{T_i} + \boldsymbol{X}_i\gamma_i + \boldsymbol{Y}_i\boldsymbol{\beta}_i + \boldsymbol{\Gamma}_i^\top \boldsymbol{f}_t + \boldsymbol{u}_i \\
&= \begin{bmatrix} \mathbf{1}_{T_i} & \boldsymbol{X}_i & \mathbf{Y}_i \end{bmatrix} \begin{pmatrix} \alpha_i \\ \gamma_i \\ \beta_i \end{pmatrix} + \boldsymbol{\Gamma}_i^\top \boldsymbol{f}_t + \boldsymbol{u}_i \\
\boldsymbol{y}_i &= \boldsymbol{Z}_i\boldsymbol{\theta}_i + \boldsymbol{\Gamma}_i^\top \boldsymbol{f}_t + \boldsymbol{u}_i
\end{aligned} \tag{35}$$

where $\boldsymbol{f}_t \in \mathbb{R}^k$ is the latent common factor signal, $\Gamma_i^\top \in \mathbb{R}^{T_i \times k}$

We approximate the unobserved factors with **cross-sectional averages**:

Define the projection matrix $\boldsymbol{M}_{\overline{\boldsymbol{Z}}_i}$ onto the orthogonal complement of the cross-sectional means:

$$\overline{\boldsymbol{Z}}_i := \begin{bmatrix} \mathbf{1} & \overline{\boldsymbol{X}}_i & \overline{\boldsymbol{Y}}_i \end{bmatrix} \tag{36}$$

$$\boldsymbol{M}_{\overline{\boldsymbol{Z}}_i} = \boldsymbol{I}_{T_i} - \overline{\boldsymbol{Z}}_i \left(\overline{\boldsymbol{Z}}_i^\top \overline{\boldsymbol{Z}}_i \right)^{-1} \overline{\boldsymbol{Z}}_i^\top \tag{37}$$

Then, apply the projection matrix $\boldsymbol{M}_{\overline{\boldsymbol{Z}}_i}$

$$\begin{aligned}
\boldsymbol{M}_{\overline{\boldsymbol{Z}}_i}\boldsymbol{y}_i = \alpha_i \boldsymbol{M}_{\overline{\boldsymbol{Z}}_i}\mathbf{1}_k + \boldsymbol{M}_{\overline{\boldsymbol{Z}}_i}\boldsymbol{Z}_i\boldsymbol{\theta}_i + \boldsymbol{M}_{\overline{\boldsymbol{Z}}_i}\boldsymbol{\Gamma}_i^\top \boldsymbol{f}_t + \boldsymbol{M}_{\overline{\boldsymbol{Z}}_i}\boldsymbol{u}_i \\
\tilde{\boldsymbol{y}}_i = \tilde{\boldsymbol{Z}}_i\boldsymbol{\theta}_i + \tilde{\boldsymbol{u}}_i
\end{aligned} \tag{38}$$

Assuming that $\boldsymbol{u}_i$ is orthogonal to the column space of $\overline{\boldsymbol{Z}}_i$, $\tilde{\boldsymbol{u}}_i \approx \boldsymbol{u}_i$.

Appendix F: Expectation and Variance of the Wald Statistic

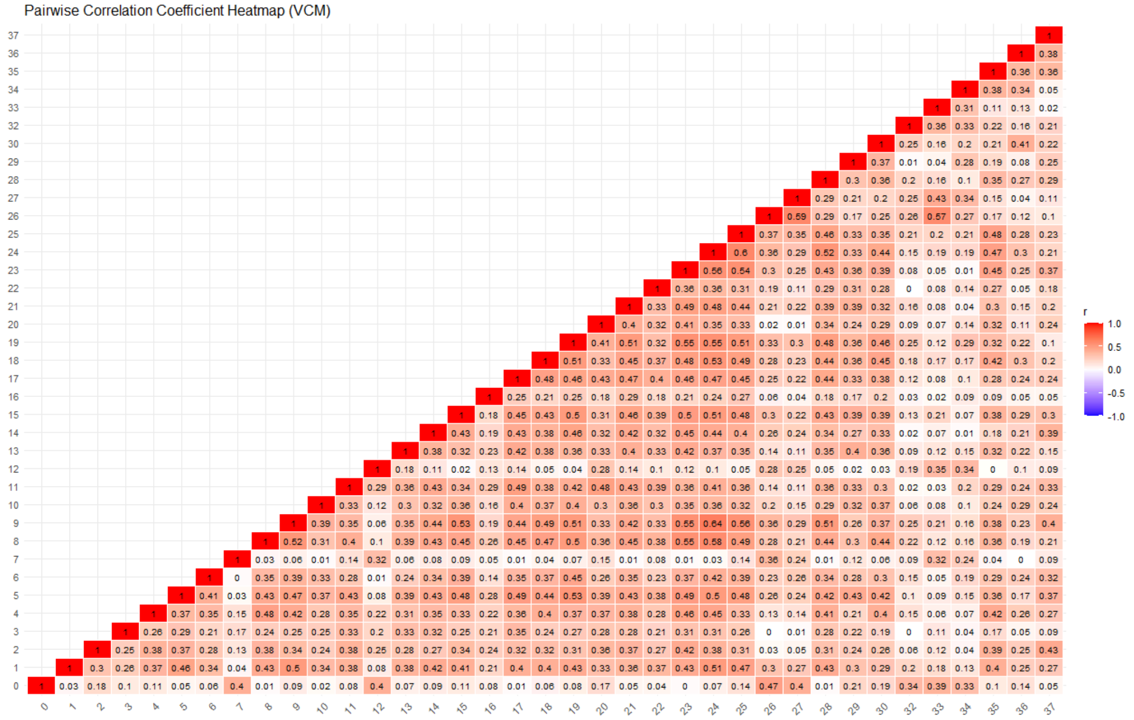
$$\mathbb{E}\Big(W_{i,T_i}\Big) = K \times \frac{T_i - 2K - 1}{T_i - 2K - 3} \tag{39}$$

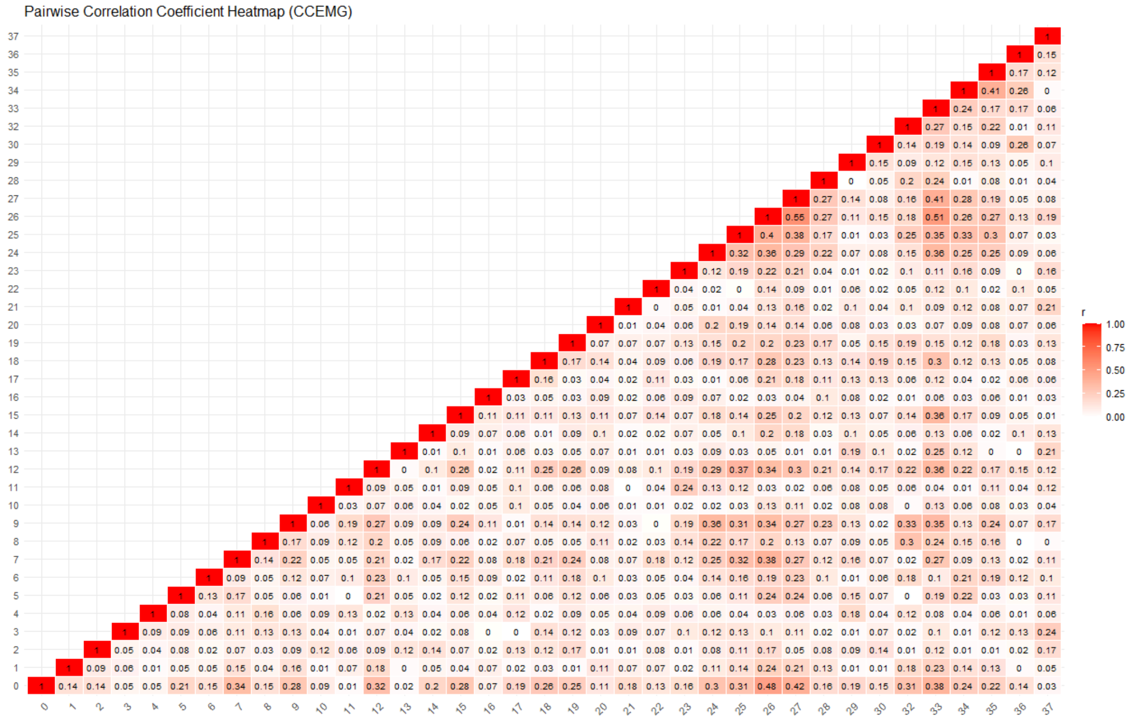
$$\begin{aligned} \mathbb{E}\Big(W_{i,T_i}^2\Big) &= \mathbb{V}ar\Big(W_{i,T_i}\Big) + \mathbb{E}\Big(W_{i,T_i}\Big)^2 \\ &= 2K \times \frac{(T_i - 2K - 1)^2}{(T_i - 2K_i - 3)^2} \times \frac{T_i - K - 3}{T_i - 2K - 5} + K^2 \times \frac{(T_i - 2K - 1)^2}{(T_i - 2K - 3)^2} \end{aligned} \tag{40}$$

$\mathbb{E}\Big(W_{i,T_i}\Big) \longrightarrow K$ as $\min\{T_i\} \longrightarrow \infty$

$\mathbb{E}\Big(W_{i,T_i}^2\Big) \longrightarrow 2K + K^2$ as $\min\{T_i\} \longrightarrow \infty$

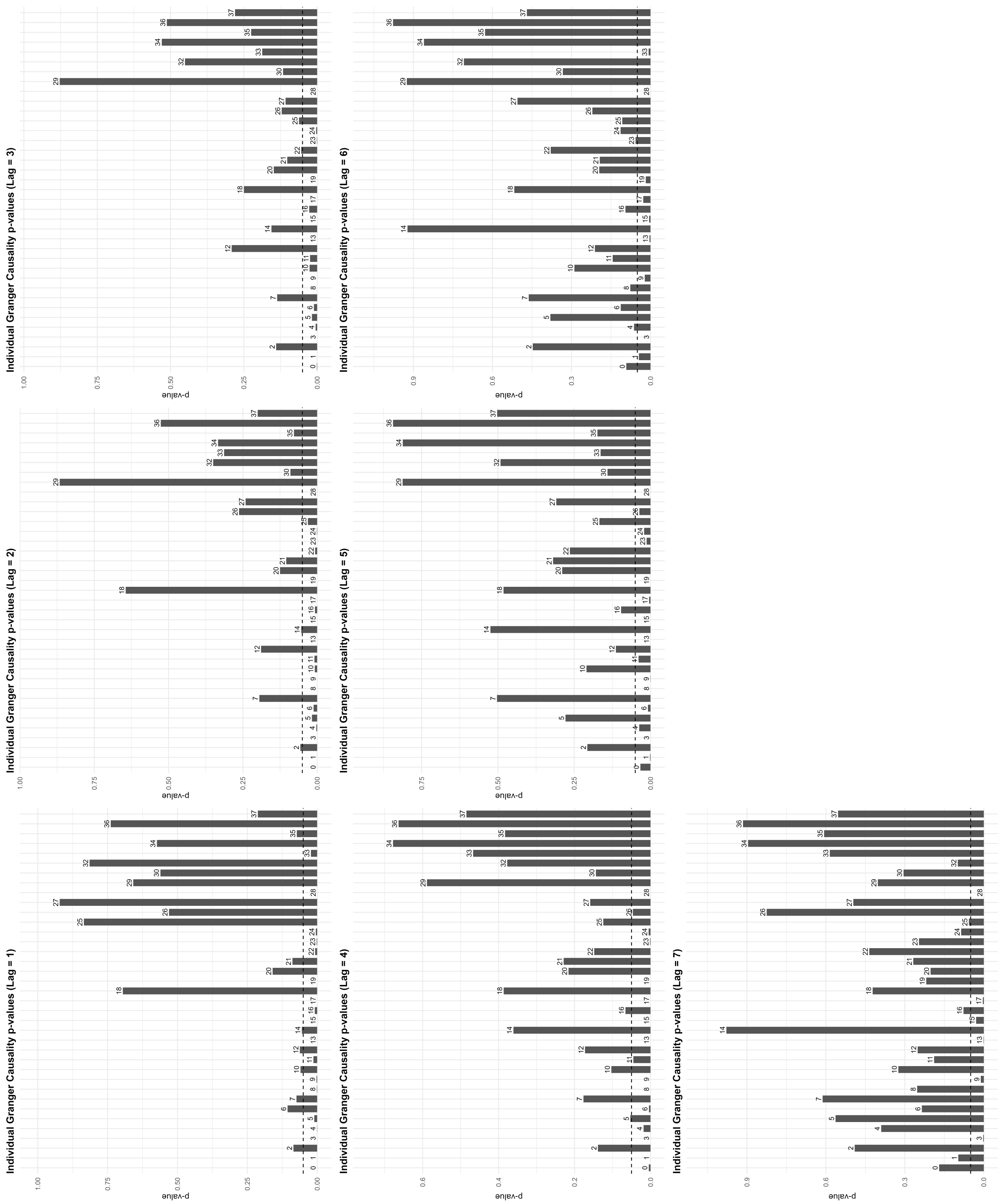
Appendix G: Pairwise correlation coefficient matrices for store- and drink- panels

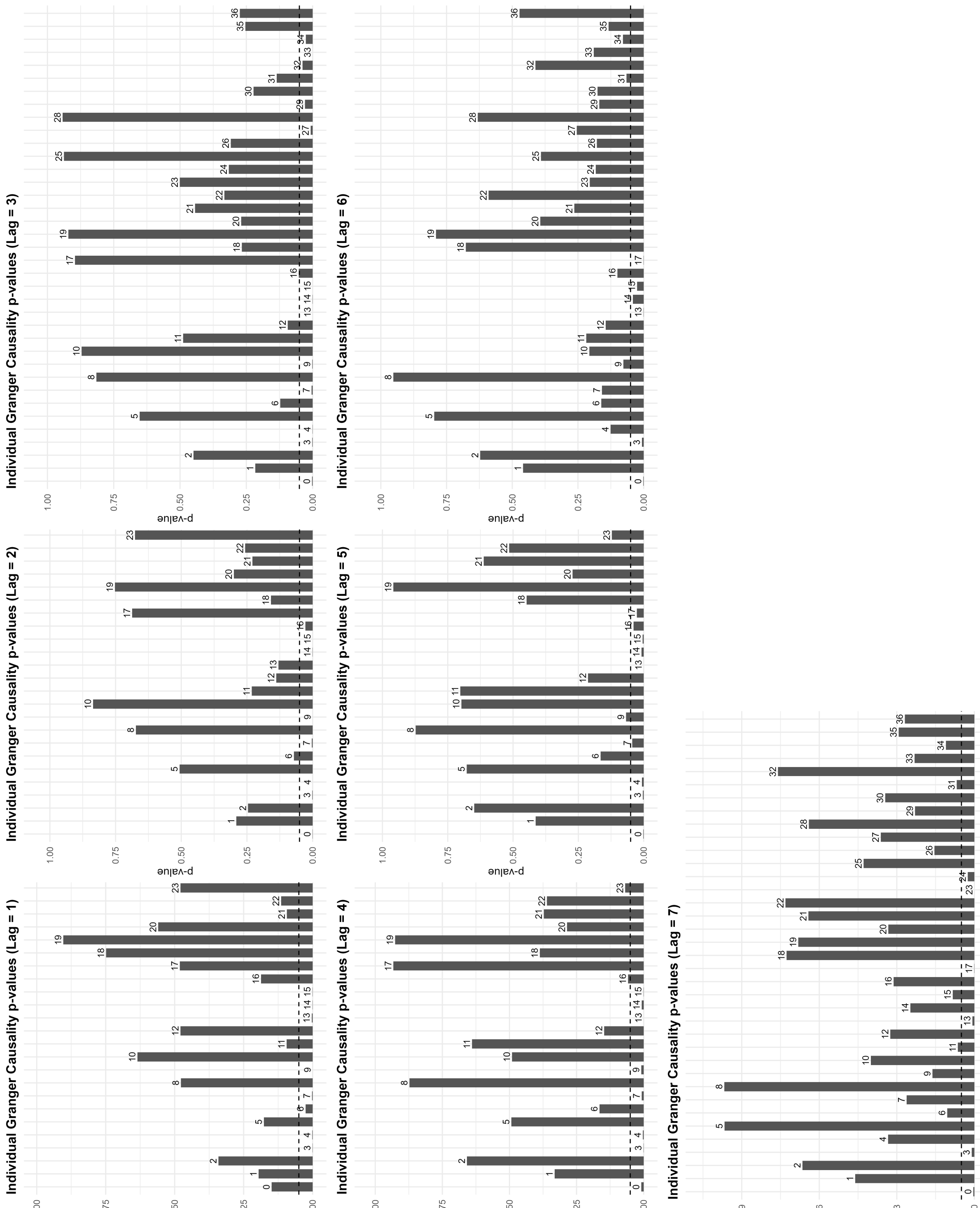




Appendix H : HNC individual p-value visualisation

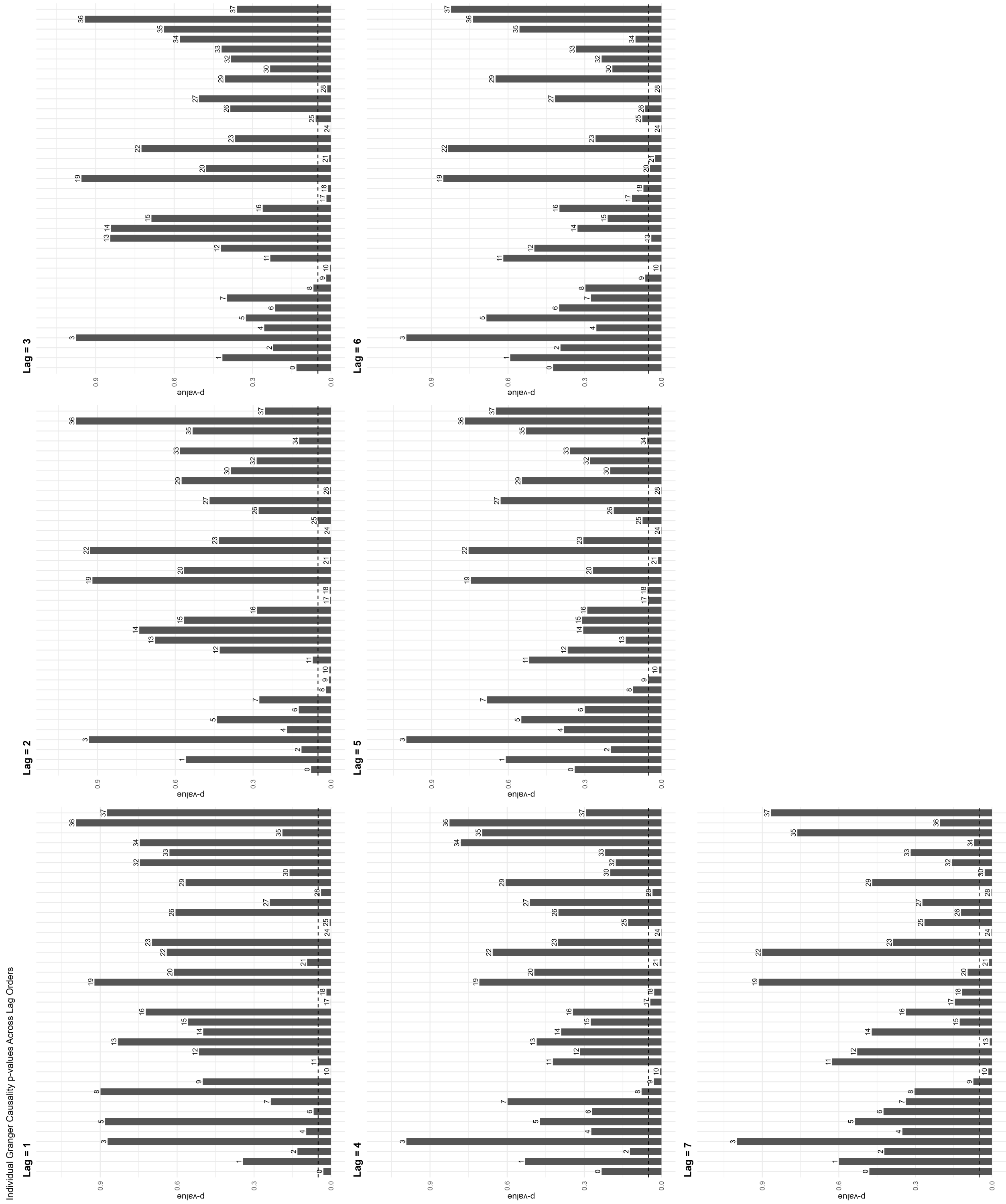
Grid of individual Granger causality HNC test p-values on revenue at the store and drink level respectively, across all lag orders for store panels without projecting onto the orthogonal component of the common factors. The dotted line indicates the $\alpha = 0.05$ sig. level.

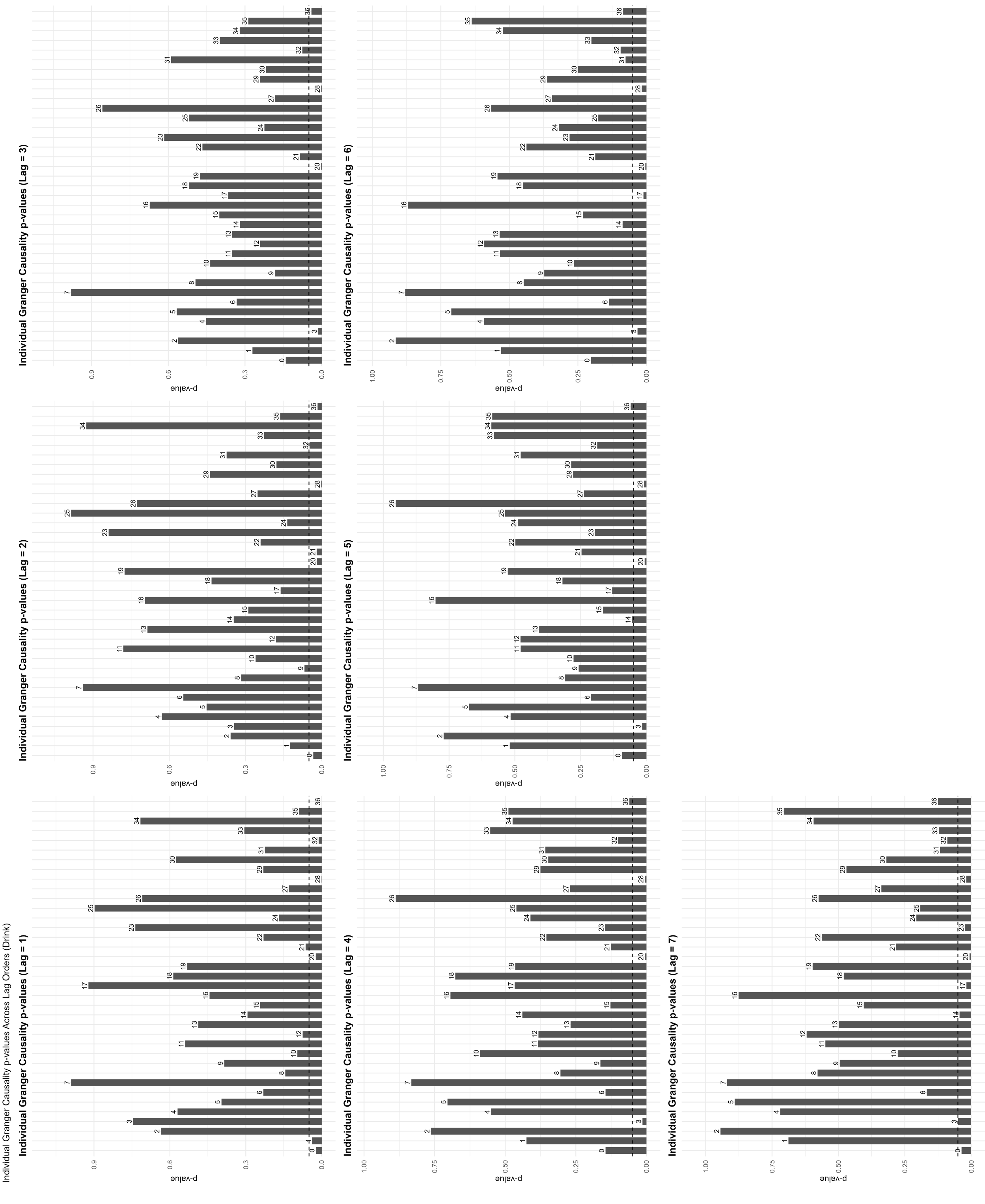



Individual Granger Causality p-values (Lag = 1)
Individual Granger Causality p-values (Lag = 2)
Individual Granger Causality p-values (Lag = 3)
Individual Granger Causality p-values (Lag = 4)
Individual Granger Causality p-values (Lag = 5)
Individual Granger Causality p-values (Lag = 6)
Individual Granger Causality p-values (Lag = 7)
p-value

Appendix I: Corrected HNC individual p-value visualisation

Grid of individual Granger causality p-values across all lag orders for Store panels (left) and Drink panels (right), on the projected data as in Equation 18. The dashed line represents the $\alpha = 0.05$ significance threshold.




Individual Granger Causality p-values Across Lag Orders (Drink)
Individual Granger Causality p-values (Lag = 1)
Individual Granger Causality p-values (Lag = 2)
Individual Granger Causality p-values (Lag = 3)
Individual Granger Causality p-values (Lag = 4)
Individual Granger Causality p-values (Lag = 5)
Individual Granger Causality p-values (Lag = 6)
Individual Granger Causality p-values (Lag = 7)
p-value